GEOMETRY · COMMON CORE HIGH SCHOOL MATH

Question Collection

Main • Backward Branches • Forward 1 • Forward 2 (later courses)

96 question sets • 2915 questions

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Precise definitions of angle, circle, and line segment

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 1	Identify points, lines, rays, segments, and angles	**4.G.A.1**
BACKWARD 2	See an angle as two rays with a common endpoint, measured by…	**4.MD.C.5.a**
BACKWARD 3	Use the center, radius, and diameter of a circle	**7.G.B.4**
FORWARD 1	Test proposed definitions with counterexamples	**HSG-CO.A.1**
FORWARD 2	Write precise definitions of new figures from earlier definitions	**HSG-CO.A.1**

SET 1 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of a circle. Use only the words point, plane, and distance, and words built from them.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q1/5

Points A and B lie on line ℓ. Which is a precise definition of segment AB?

A. Every point of line ℓ

B. Points A and B together with every point C of line ℓ for which AC + CB = AB

C. Point A and every point of line ℓ on the same side of A as B

D. Every point of the plane that is closer to A than to B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q2/5

Write a precise definition of an angle. Name its vertex and its sides in your definition.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q3/5

Mia writes: "A circle is all the points that are 5 cm from point O."

What must she add to make the definition precise? Explain why.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q4/5

True or false?

An angle with its vertex at the center of a circle that cuts off $\frac{1}{360}$ of the circle measures 1°, whatever the radius of the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • MAIN • Q5/5

SET 1 • HSG-CO.A.1

BACKWARD 1

Identify points, lines, rays, segments, and angles • Grade 4 • Geometry • 4.G.A.1

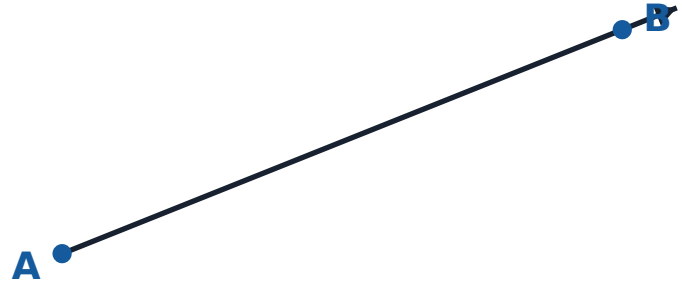
What is shown in the figure?

A. Ray AB

B. Ray BA

C. Line AB

D. Segment AB



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q1/5

How many endpoints does a line segment have? How many does a ray have?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q2/5

True or false?

A line segment has exactly two endpoints.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q3/5

Which figure goes on without end in exactly one direction?

A. A line

B. A point

C. A line segment

D. A ray

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q4/5

True or false?

Ray AB and ray BA are the same ray.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 1 • BACKWARD 1 • Q5/5

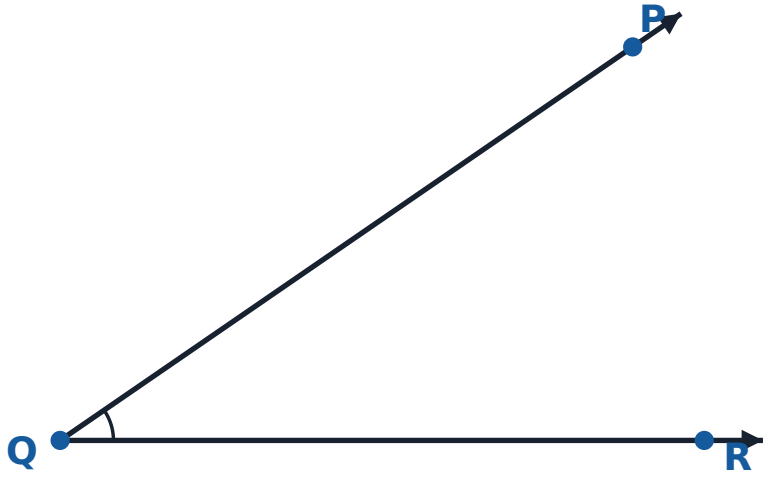
SET 1 • HSG-CO.A.1

BACKWARD 2

See an angle as two rays with a common endpoint, measured by an arc • Grade 4 • Measurement & Data • 4.MD.C.5.a

Name the vertex and the two sides of ∠PQR.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q1/5

An angle has its vertex at the center of a circle and cuts off $\frac{1}{4}$ of the circle. What is its measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q2/5

An angle measures 60°. What fraction of a circle centered at its vertex does it cut off?

Fraction: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q3/5

True or false?

The measure of an angle depends on how long its sides are drawn.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q4/5

True or false?

A one-degree angle turns through $\frac{1}{360}$ of a full circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.a • Set 1 • BACKWARD 2 • Q5/5

SET 1 • HSG-CO.A.1

BACKWARD 3

Use the center, radius, and diameter of a circle • Grade 7 • Geometry • 7.G.B.4

A circle has center O and radius 7 cm. Point P is in the plane of the circle, 7 cm from O. Is P on the circle? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q1/5

A circle has a diameter of 18 in. What is its radius?

Radius: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q2/5

A circle has center C and radius 4. Point D is in the same plane, 3 units from C. Where is D?

A. Inside the circle

B. Outside the circle

C. On the circle

D. There is not enough information

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q3/5

True or false?

Every radius of a circle has the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q4/5

True or false?

A point 10 units from the center of a circle of radius 5 lies on the circle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-B3-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 1 • BACKWARD 3 • Q5/5

SET 1 • HSG-CO.A.1

FORWARD 1

Test proposed definitions with counterexamples • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Proposed definition: "A circle is a set of points that are all the same distance from a point O."

Give a set that fits this wording but is NOT a circle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q1/5

Proposed definition: "Point M is the midpoint of segment AB if AM = MB."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q2/5

Which set fits the description "all points 2 units from point O" but is NOT a circle?

A. A circle of radius 2 centered at O in a plane

B. A sphere of radius 2 centered at O

C. A disk of radius 2

D. A segment of length 2 from O

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q3/5

True or false?

"Two lines are parallel if they never meet" is a complete definition, even in three-dimensional space.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q4/5

Proposed definition: "A diameter is a segment that passes through the center of a circle."

Give a counterexample, and repair the definition.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 1 • Q5/5

SET 1 • HSG-CO.A.1

FORWARD 2

Write precise definitions of new figures from earlier definitions • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of ray AB. Use the word "between".

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q1/5

Write a precise definition of the midpoint of segment AB.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q2/5

Segment PQ is a diameter of a circle with center O. Write a precise definition of a semicircle with endpoints P and Q.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q3/5

Which is a precise definition of the bisector of ∠ABC?

A. A ray from B that lies outside ∠ABC

B. The segment that joins A and C

C. Any line through B

D. Ray BD, where D is in the interior of ∠ABC and m∠ABD = m∠DBC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q4/5

True or false?

The definition "the midpoint of AB is a point M with AM = MB" is precise enough, because every such point lies on segment AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S1-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 1 • FORWARD 2 • Q5/5

Precise definitions of perpendicular lines and parallel lines

MAIN	Geometry main question	**HSG-CO.A.1**
BACKWARD 1	Recognize perpendicular and parallel lines and right angles	**4.G.A.1**
BACKWARD 2	Find the other angles when two lines cross	**7.G.B.5**
FORWARD 1	Prove consequences of the definition of perpendicular lines	**HSG-CO.C.9**
FORWARD 2	Reason about three lines in a plane and in space	**HSG-CO.C.9**

SET 2 • HSG-CO.A.1

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Write a precise definition of parallel lines.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q1/5

Write a precise definition of perpendicular lines.

Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q2/5

Lines ℓ and m do not intersect. Which additional fact makes them parallel?

A. They lie in the same plane

B. They both look horizontal in a drawing

C. They are drawn the same length

D. They are both perpendicular to one segment in space

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q3/5

True or false?

If two lines meet and form one right angle, they form four right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

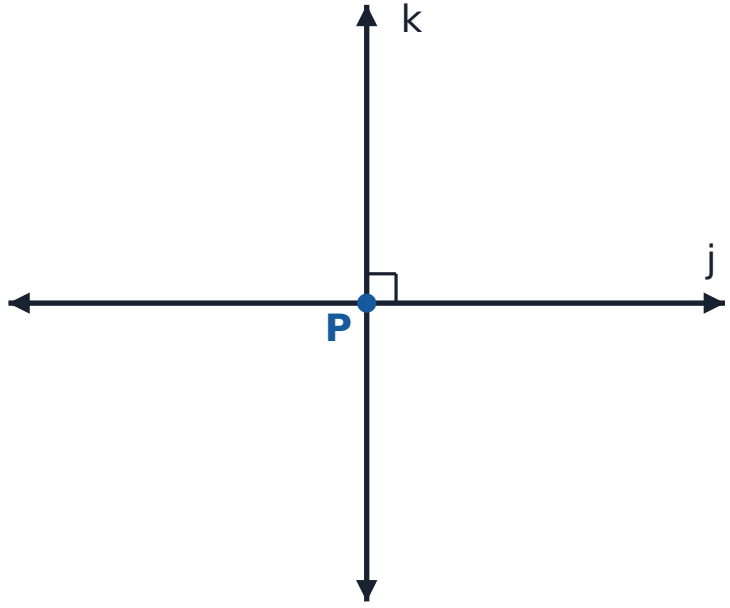
ID S2-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q4/5

Lines j and k meet at point P, and one of the angles they form is marked as a right angle.

Are j and k perpendicular? Use the definition to explain.

Answer: _____



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 2 • MAIN • Q5/5

SET 2 • HSG-CO.A.1

BACKWARD 1

Recognize perpendicular and parallel lines and right angles • Grade 4 • Geometry • 4.G.A.1

True or false?

Lines that meet at a 90° angle are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q1/5

True or false?

Any two lines that cross are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q2/5

Two perpendicular lines cross. How many right angles do they form?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q3/5

What is true of the opposite sides of a rectangle?

A. They are parallel

B. They are perpendicular

C. They meet at one point

D. They are neither parallel nor perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q4/5

Which pair of streets is most like perpendicular lines?

A. Two straight streets that never meet

B. Two streets that curve around a park

C. Two streets that meet at a sharp point

D. Two streets that cross at a square corner

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B1-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 2 • BACKWARD 1 • Q5/5

SET 2 • HSG-CO.A.1

BACKWARD 2

Find the other angles when two lines cross • Grade 7 • Geometry • 7.G.B.5

Two lines cross. One of the angles measures 90°. Find the other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q1/5

Two lines cross. One of the angles measures 70°. Find the other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q2/5

True or false?

Vertical angles formed by two crossing lines are equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q3/5

Two angles form a linear pair. One measures 90°. What does the other measure?

A. 180°

B. 45°

C. 90°

D. 270°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q4/5

True or false?

Two lines cross and one angle measures 100°. The angle next to it also measures 100°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-B2-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 2 • BACKWARD 2 • Q5/5

SET 2 • HSG-CO.A.1

FORWARD 1

Prove consequences of the definition of perpendicular lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are perpendicular and meet at point P. Prove that all four angles at P are right angles.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q1/5

Two lines meet at point P and form a linear pair of congruent angles. Prove that the lines are perpendicular.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q2/5

Lines a and b intersect, and one of the angles they form measures (3x + 15)°. For what value of x are a and b perpendicular?

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q3/5

True or false?

If two intersecting lines form a pair of vertical angles that are supplementary, the lines are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q4/5

Lines p and q meet at point O. Which statement is enough to prove that p ⊥ q?

A. Two of the angles at O are vertical angles

B. Two adjacent angles at O add to 180°

C. Two of the angles at O are acute

D. One of the four angles at O measures 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 1 • Q5/5

SET 2 • HSG-CO.A.1

FORWARD 2

Reason about three lines in a plane and in space • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

In a plane, lines a and b are both perpendicular to line t. Prove that a and b are parallel.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q1/5

True or false?

In three-dimensional space, two lines that are each perpendicular to the same line must be parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q2/5

In a plane, line a is parallel to line b, and line t is perpendicular to line a. What is true about t and b?

A. t and b do not meet

B. t is perpendicular to b

C. t is parallel to b

D. The angle between t and b cannot be known

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q3/5

In a coordinate plane, line t is the x-axis. Line a is perpendicular to t at (0, 0), and line b is perpendicular to t and passes through (5, 2). Write an equation of line b.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q4/5

True or false?

In a plane, through a point not on a line there is exactly one line parallel to the given line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S2-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 2 • FORWARD 2 • Q5/5

Transformations as functions: inputs, outputs, and preimages

MAIN	Geometry main question	**HSG-CO.A.2**
BACKWARD 1	Evaluate a rule for a given input	**8.F.A.1**
BACKWARD 2	Apply coordinate rules for translations, reflections, and rotations	**8.G.A.3**
BACKWARD 3	Read and locate ordered pairs in all four quadrants	**6.NS.C.6.c**
FORWARD 1	Compose two transformations and write one rule	**HSG-CO.A.2**
FORWARD 2	Find the inverse of a transformation	**HSG-CO.A.2**

SET 3 • HSG-CO.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

The transformation T is defined by T(x, y) = (x + 4, y − 3). Find T(2, 5).

T(2, 5) = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q1/5

The transformation R is defined by $R(x, y) = (-y, x)$. Find $R(3, -1)$.

$R(3, -1) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q2/5

The transformation F takes each point (x, y) to (x, −y). The output point is F(P) = (4, 7). Find the input point P.

P = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q3/5

Every point of the plane is moved 5 units right and 2 units up. Which function describes this transformation?

A. $T(x, y) = (x - 5, y - 2)$

B. $T(x, y) = (5x, 2y)$

C. $T(x, y) = (x + 2, y + 5)$

D. $T(x, y) = (x + 5, y + 2)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q4/5

True or false?

Under the transformation D(x, y) = (2x, 2y), the point (0, 0) is its own output.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • MAIN • Q5/5

SET 3 • HSG-CO.A.2

BACKWARD 1

Evaluate a rule for a given input • Grade 8 • Functions • 8.F.A.1

$f(x) = 3x - 2$. Find $f(4)$.

$f(4) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q1

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q1/5

A rule takes a number, doubles it, and adds 1. What is the output for the input 6?

Output: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q2

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q2/5

$g(x) = x + 5$. Which input gives the output 12?

Input: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q3

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q3/5

True or false?

A function assigns exactly one output to each input.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q4

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q4/5

True or false?

The rule "the input 3 gives the outputs 4 and 5" describes a function.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B1-Q5

Grade 8 • Functions • 8.F.A.1 • Set 3 • BACKWARD 1 • Q5/5

SET 3 • HSG-CO.A.2

BACKWARD 2

Apply coordinate rules for translations, reflections, and rotations • Grade 8 • Geometry • 8.G.A.3

Translate the point (2, −1) 3 units left and 4 units up. Where does it go?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q1/5

Reflect the point (5, 2) across the y-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q2/5

Rotate the point (4, 1) by 180° about the origin.

A. (1, 4)

B. (−1, 4)

C. (4, −1)

D. (−4, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q3/5

True or false?

Reflecting the point (3, 6) across the x-axis gives (3, −6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q4/5

True or false?

Rotating the point (0, 5) by 90° counterclockwise about the origin gives (5, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B2-Q5

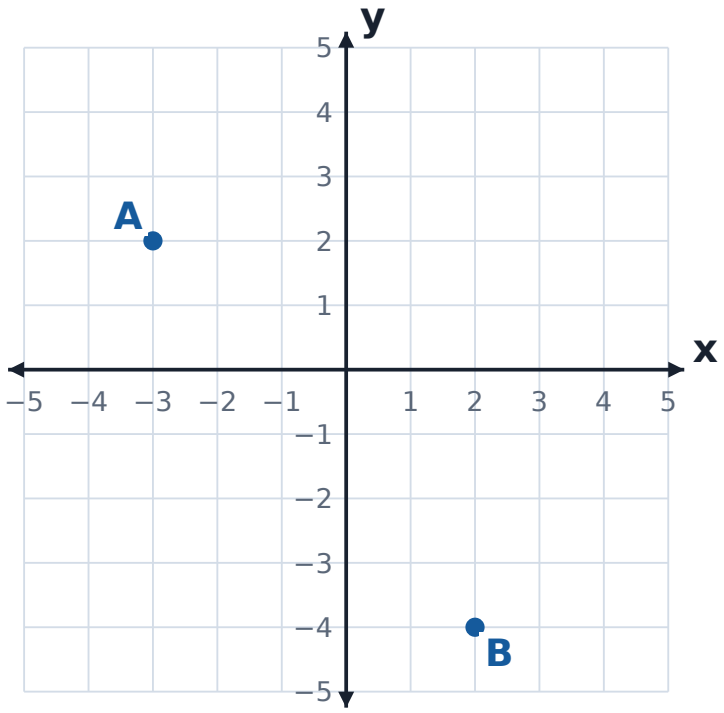
Grade 8 • Geometry • 8.G.A.3 • Set 3 • BACKWARD 2 • Q5/5

SET 3 • HSG-CO.A.2

BACKWARD 3

Read and locate ordered pairs in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

What are the coordinates of point A?



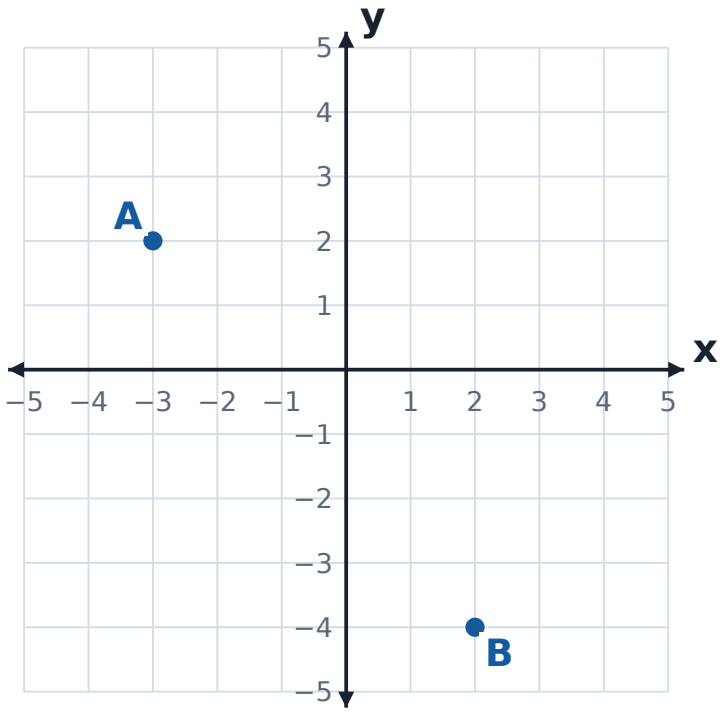
A = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q1/5

What are the coordinates of point B?



B = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q2/5

Which point is in Quadrant III?

A. (−2, −5)

B. (−2, 5)

C. (2, 5)

D. (2, −5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q3/5

True or false?

The point (0, −3) lies on the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q4/5

True or false?

The point $(-2, 5)$ is 2 units to the right of the y-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-B3-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 3 • BACKWARD 3 • Q5/5

SET 3 • HSG-CO.A.2

FORWARD 1

Compose two transformations and write one rule • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

$T(x, y) = (x + 2, y)$ and $R(x, y) = (-x, y)$. Find $R(T(1, 4))$.

$R(T(1, 4)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q1/5

$T(x, y) = (x + 2, y)$ and $R(x, y) = (-x, y)$. Write a single rule for $R(T(x, y))$.

$R(T(x, y)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q2/5

$T(x, y) = (x + 2, y)$ and $R(x, y) = (-x, y)$. Find $T(R(1, 4))$.

$T(R(1, 4)) =$ _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q3/5

True or false?

For any two transformations, doing them in either order gives the same result.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q4/5

$T(x, y) = (x + 3, y - 1)$ and $S(x, y) = (x - 1, y + 5)$. Which rule is $S(T(x, y))$?

A. $(x + 2, y + 4)$

B. $(x - 3, y - 5)$

C. $(x + 2, y - 4)$

D. $(x + 4, y - 6)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 1 • Q5/5

SET 3 • HSG-CO.A.2

FORWARD 2

Find the inverse of a transformation • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

Find a rule for the inverse of $T(x, y) = (x - 6, y + 2)$.

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q1/5

$R(x, y) = (-y, x)$ is a 90° counterclockwise rotation about the origin. Find a rule for its inverse.

Inverse: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q2/5

True or false?

Every rotation is its own inverse.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q3/5

D(x, y) = (3x, 3y). Which rule undoes D?

A. $(x - 3, y - 3)$

B. $(3y, 3x)$

C. $(-3x, -3y)$

D. $(\frac{x}{3}, \frac{y}{3})$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q4/5

True or false?

The transformation P(x, y) = (x, 0), which moves every point straight up or down to the x-axis, has an inverse transformation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S3-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 3 • FORWARD 2 • Q5/5

Compare transformations that preserve distance and angle with those that do not

MAIN	Geometry main question	**HSG-CO.A.2**
BACKWARD 1	Find the distance between two points	**8.G.B.8**
BACKWARD 2	Know that rigid motions keep lengths and angle measures	**8.G.A.1**
FORWARD 1	Classify transformations as rigid motions, similarity transformatio...	**HSG-SRT.A.2**
FORWARD 2	Prove with variables that a transformation preserves or scales ev...	**HSG-SRT.A.2**

SET 4 • HSG-CO.A.2

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.2

Which transformation does NOT preserve distance?

A. $R(x, y) = (-x, y)$

B. $T(x, y) = (x + 4, y - 1)$

C. $H(x, y) = (2x, y)$

D. $Q(x, y) = (-y, x)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q1/5

H(x, y) = (2x, y). Points A(0, 0) and B(3, 4) are 5 units apart.

Find H(A), H(B), and the distance between them. Does H preserve distance?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q2/5

The translation $T(x, y) = (x - 2, y + 5)$ maps $A(1, 1)$ and $B(4, 5)$ to A' and B'. Show that $A'B' = AB$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q3/5

True or false?

The transformation V(x, y) = (x, 3y) preserves the measure of every angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q4/5

S(x, y) = (3x, 3y). Does S preserve distance? Does S preserve angle measure? Explain each.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.2 • Set 4 • MAIN • Q5/5

SET 4 • HSG-CO.A.2

BACKWARD 1

Find the distance between two points • Grade 8 • Geometry • 8.G.B.8

Find the distance between (1, 2) and (4, 6).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q1/5

Find the distance between $(-2, 3)$ and $(4, 3)$.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q2/5

Find the distance between (0, 0) and (5, 12).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q3/5

True or false?

The distance from (0, 0) to (1, 1) is 2.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q4/5

Find the distance between $(2, -1)$ and $(5, 3)$.

A. 25

B. $\sqrt{7}$

C. 5

D. 7

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B1-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 4 • BACKWARD 1 • Q5/5

SET 4 • HSG-CO.A.2

BACKWARD 2

Know that rigid motions keep lengths and angle measures • Grade 8 • Geometry • 8.G.A.1

True or false?

A rotation keeps every segment the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q1/5

True or false?

A reflection can change the measure of an angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q2/5

A triangle is translated. What changes?

A. Its angle measures

B. Its side lengths

C. The positions of its vertices

D. Its area

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q3/5

A segment 7 cm long is reflected across a line. How long is its image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q4/5

A 35° angle is rotated 90° about a point. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 4 • BACKWARD 2 • Q5/5

SET 4 • HSG-CO.A.2

FORWARD 1

Classify transformations as rigid motions, similarity transformations, or neither • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Which transformation preserves angle measure but not distance?

A. $(x, y) \rightarrow (4x, y)$

B. $(x, y) \rightarrow (x + 4, y)$

C. $(x, y) \rightarrow (x, y + 4)$

D. $(x, y) \rightarrow (4x, 4y)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q1/5

Classify $(x, y) \to (-2x, -2y)$ as a rigid motion, a similarity transformation that is not rigid, or neither. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q2/5

Classify $(x, y) \to (x + y, y)$ as a rigid motion, a similarity transformation that is not rigid, or neither. Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q3/5

True or false?

Every similarity transformation is a rigid motion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q4/5

True or false?

A transformation that multiplies every distance by the same number k > 0 always changes angle measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 1 • Q5/5

SET 4 • HSG-CO.A.2

FORWARD 2

Prove with variables that a transformation preserves or scales every distance • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Prove that $T(x, y) = (x + a, y + b)$ preserves the distance between ANY two points (x_1, y_1) and (x_2, y_2).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q1/5

Prove that D(x, y) = (3x, 3y) multiplies the distance between ANY two points by 3.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q2/5

$H(x, y) = (2x, y)$. The distance between the images of (x_1, y_1) and (x_2, y_2) is $\sqrt{(4(x_2 - x_1)^2 + (y_2 - y_1)^2)}$.

For which pairs of points is the distance unchanged?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q3/5

True or false?

One example in which a distance stays the same is enough to prove that a transformation preserves every distance.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q4/5

True or false?

One example in which a distance is preserved is enough to prove that a transformation preserves distance.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S4-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 4 • FORWARD 2 • Q5/5

Rotations and reflections that carry a rectangle, parallelogram, or trapezoid onto itself

MAIN	Geometry main question	**HSG-CO.A.3**
BACKWARD 1	Identify lines of symmetry of figures	**4.G.A.3**
BACKWARD 2	Know the side and angle properties of parallelograms, rectangles,…	**5.G.B.3**
BACKWARD 3	Rotate points by 90° and 180° about the origin	**8.G.A.3**
FORWARD 1	Use a symmetry to deduce congruent parts of the figure	**HSG-CO.B.6**
FORWARD 2	Combine symmetries and identify a figure from its symmetries	**HSG-CO.A.5**

SET 5 • HSG-CO.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

Rectangle ABCD is not a square. Describe every rotation (with angle greater than 0° and at most 360°) and every reflection that carries the rectangle onto itself.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q1/5

Parallelogram PQRS has no right angles, and its sides are not all the same length. Which transformation carries it onto itself?

A. A 90° rotation about the point where its diagonals meet

B. A reflection across diagonal PR

C. A reflection across the line through the midpoints of PQ and RS

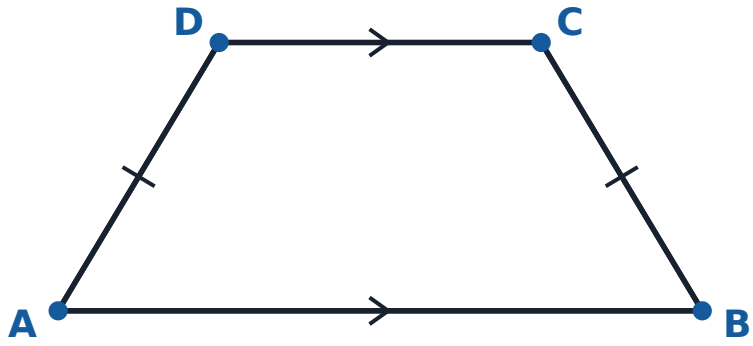
D. A 180° rotation about the point where its diagonals meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q2/5

Trapezoid ABCD is isosceles, with bases AB and DC and legs AD = BC. Describe every rotation and reflection that carries it onto itself.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q3/5

True or false?

A 90° rotation about its center carries a rectangle that is not a square onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q4/5

Trapezoid EFGH has EF ∥ HG, EF = 4, HG = 8, and legs FG = 5 and EH = 3. Which rotations and reflections carry it onto itself?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 5 • MAIN • Q5/5

SET 5 • HSG-CO.A.3

BACKWARD 1

Identify lines of symmetry of figures • Grade 4 • Geometry • 4.G.A.3

How many lines of symmetry does a square have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q1/5

How many lines of symmetry does a rectangle that is not a square have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q2/5

True or false?

Each diagonal of a rectangle that is not a square is a line of symmetry of the rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q3/5

Which figure has exactly one line of symmetry?

A. An isosceles triangle that is not equilateral

B. A square

C. A circle

D. A scalene triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q4/5

True or false?

An isosceles trapezoid has a line of symmetry.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B1-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 5 • BACKWARD 1 • Q5/5

SET 5 • HSG-CO.A.3

BACKWARD 2

Know the side and angle properties of parallelograms, rectangles, and trapezoids • Grade 5 • Geometry • 5.G.B.3

True or false?

In every parallelogram, the diagonals have equal lengths.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q1/5

True or false?

Every rectangle is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q2/5

Which property do all rectangles have?

A. Diagonals that are perpendicular

B. Four sides of equal length

C. Four right angles

D. Exactly one pair of parallel sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q3/5

A parallelogram has one angle of 70°. Find its other three angles.

Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q4/5

An isosceles trapezoid has a 65° angle at one end of its longer base. What is the angle at the other end of that base?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 5 • BACKWARD 2 • Q5/5

SET 5 • HSG-CO.A.3

BACKWARD 3

Rotate points by 90° and 180° about the origin • Grade 8 • Geometry • 8.G.A.3

Rotate the point (3, 1) by 180° about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q1/5

Rotate the point (2, 0) by 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q2/5

True or false?

A 180° rotation about the origin sends (x, y) to (−x, −y).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q3/5

Rotate the point $(-4, 2)$ by 90° counterclockwise about the origin.

A. $(4, -2)$

B. $(-4, -2)$

C. $(2, 4)$

D. $(-2, -4)$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q4/5

True or false?

A rotation of 360° about a point moves every point to a new position.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 5 • BACKWARD 3 • Q5/5

SET 5 • HSG-CO.A.3

FORWARD 1

Use a symmetry to deduce congruent parts of the figure • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Isosceles trapezoid ABCD (AB ∥ DC) is carried onto itself by the reflection across the line through the midpoints of AB and DC. Which vertex does A go to? Use this to name a pair of congruent angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q1/5

A 180° rotation about point O carries parallelogram PQRS onto itself, sending P to R and Q to S. Name two pairs of congruent sides that follow.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q2/5

True or false?

Because a reflection carries an isosceles trapezoid onto itself, its two diagonals are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q3/5

A reflection carries rectangle ABCD onto itself and swaps vertices A and B (and C and D). Which equality follows?

A. AB = BC

B. AC = AB

C. AD = AB

D. AD = BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q4/5

True or false?

If a 180° rotation about a point carries a quadrilateral onto itself, the quadrilateral must be a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 5 • FORWARD 1 • Q5/5

SET 5 • HSG-CO.A.3

FORWARD 2

Combine symmetries and identify a figure from its symmetries • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Rectangle ABCD (not a square) has center O. Reflect it across the line through the midpoints of AB and CD, then across the line through the midpoints of BC and DA. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q1/5

A quadrilateral is carried onto itself by exactly two transformations: the 360° rotation and one reflection across a line through two of its vertices. Which could it be?

A. A parallelogram that is not a rectangle or a rhombus

B. A square

C. A rectangle that is not a square

D. A kite that is not a rhombus

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q2/5

True or false?

A quadrilateral that is carried onto itself by a 180° rotation about a point must be a rectangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q3/5

A quadrilateral is carried onto itself by a 180° rotation but by no reflection. What is the most specific name for it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q4/5

True or false?

Doing the same reflection twice gives the same result as the 360° rotation (the identity).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S5-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 5 • FORWARD 2 • Q5/5

Rotations and reflections that carry a regular polygon onto itself

MAIN	Geometry main question	**HSG-CO.A.3**
BACKWARD 1	Find the angle of a fraction of a full turn	**4.MD.C.5.b**
BACKWARD 2	Recognize regular polygons	**5.G.B.3**
BACKWARD 3	Find lines of symmetry of regular polygons	**4.G.A.3**
FORWARD 1	Work backward from a symmetry to the regular polygon	**HSG-CO.A.3**
FORWARD 2	Combine symmetries of a regular polygon	**HSG-CO.A.5**

SET 6 • HSG-CO.A.3

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

Describe every rotation about its center, with angle greater than 0° and at most 360°, that carries a regular pentagon onto itself.

Rotations: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q1/5

How many reflections carry a regular hexagon onto itself? Describe their lines.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q2/5

Which rotation about its center carries a regular octagon onto itself?

A. 100°

B. 135°

C. 150°

D. 60°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q3/5

True or false?

Each line of symmetry of a regular nonagon (9 sides) passes through one vertex and the midpoint of the opposite side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q4/5

Describe every reflection that carries an equilateral triangle onto itself.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • MAIN • Q5/5

SET 6 • HSG-CO.A.3

BACKWARD 1

Find the angle of a fraction of a full turn • Grade 4 • Measurement & Data • 4.MD.C.5.b

How many degrees are in $\frac{1}{5}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q1/5

How many degrees are in $\frac{1}{8}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q2/5

How many degrees are in $\frac{3}{6}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q3/5

True or false?

$\frac{1}{12}$ of a full turn is 30°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q4/5

True or false?

$\frac{1}{9}$ of a full turn is 36°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 6 • BACKWARD 1 • Q5/5

SET 6 • HSG-CO.A.3

BACKWARD 2

Recognize regular polygons • Grade 5 • Geometry • 5.G.B.3

Which is a regular polygon?

A. A right triangle

B. A rhombus that is not a square

C. A rectangle that is not a square

D. An equilateral triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q1

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q1/5

True or false?

A polygon with all sides equal must be regular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q2

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q2/5

How many sides does a regular decagon have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q3

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q3/5

What is the name of the regular polygon with 4 sides?

Name: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q4

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q4/5

True or false?

A regular hexagon has 6 equal angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B2-Q5

Grade 5 • Geometry • 5.G.B.3 • Set 6 • BACKWARD 2 • Q5/5

SET 6 • HSG-CO.A.3

BACKWARD 3

Find lines of symmetry of regular polygons • Grade 4 • Geometry • 4.G.A.3

How many lines of symmetry does an equilateral triangle have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q1

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q1/5

How many lines of symmetry does a regular pentagon have?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q2

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q2/5

True or false?

Every line of symmetry of a regular polygon passes through its center.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q3

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q3/5

A line of symmetry of a square passes through

A. two adjacent vertices

B. exactly one vertex and no midpoint

C. no point of the square

D. two opposite vertices, or the midpoints of two opposite sides

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q4

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q4/5

True or false?

A regular hexagon has exactly 3 lines of symmetry.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-B3-Q5

Grade 4 • Geometry • 4.G.A.3 • Set 6 • BACKWARD 3 • Q5/5

SET 6 • HSG-CO.A.3

FORWARD 1

Work backward from a symmetry to the regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.3

The smallest positive rotation that carries a regular polygon onto itself is 24°. How many sides does it have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q1/5

A regular polygon has 20 lines of symmetry. What is the smallest positive rotation that carries it onto itself?

Rotation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q2/5

True or false?

A regular polygon can have 50° as its smallest positive rotation onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q3/5

Which can be the smallest positive rotation that carries a regular polygon onto itself?

A. 35°

B. 70°

C. 80°

D. 40°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q4/5

Counting the 360° rotation, how many rotations and reflections in all carry a regular 12-gon onto itself?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.3 • Set 6 • FORWARD 1 • Q5/5

SET 6 • HSG-CO.A.3

FORWARD 2

Combine symmetries of a regular polygon • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Regular hexagon ABCDEF is labeled counterclockwise. It is rotated 120° counterclockwise about its center. Which vertex position does A move to?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q1/5

Regular hexagon ABCDEF is labeled counterclockwise. Rotate it 60° counterclockwise about its center, then reflect it across line AD (through the original positions of A and D). Which original vertex position does vertex B end at?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q2/5

True or false?

For a regular pentagon, a reflection followed by a different reflection that carries it onto itself is a rotation that carries it onto itself.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q3/5

Square ABCD is labeled counterclockwise. Reflect it across line AC, then rotate it 90° counterclockwise about its center. Which original vertex position does vertex B end at?

A. The position of C

B. The position of A

C. The position of D

D. The position of B

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q4/5

Two successive 72° rotations of a regular pentagon about its center equal which single transformation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S6-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 6 • FORWARD 2 • Q5/5

Definitions of reflections and translations

MAIN	Geometry main question	**HSG-CO.A.4**
BACKWARD 1	Find the midpoint of a horizontal or vertical segment	**6.NS.C.8**
BACKWARD 2	Recognize perpendicular and parallel segments on a grid	**4.G.A.1**
BACKWARD 3	Reflect and translate points with coordinate rules	**8.G.A.3**
FORWARD 1	Find lines of reflection and images in slanted, non-axis cases	**HSG-CO.A.4**
FORWARD 2	Prove from the definitions that reflections and translations keep l…	**HSG-CO.A.4**

SET 7 • HSG-CO.A.4

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Complete a precise definition. A reflection across line ℓ maps each point P not on ℓ to the point P′ such that _____. Each point on ℓ maps to _____.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q1/5

Points A and B are different. Write a precise definition of the translation that maps A to B.

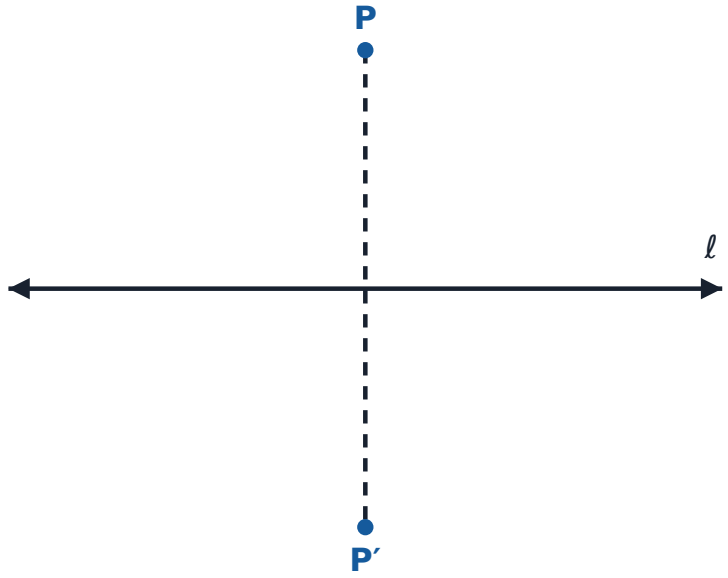
Definition: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q2/5

Point P′ is the image of point P under a reflection across line ℓ, and P is not on ℓ. Which must be true?



A. ℓ is perpendicular to PP′ and passes through its midpoint

B. ℓ passes through P

C. PP′ is parallel to ℓ

D. P and P′ are on the same side of ℓ

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q3/5

True or false?

Under a translation, all the segments that join points to their images are parallel (or on one line) and have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q4/5

A reflection maps P(1, 2) to P′(5, 2). Use the definition to find the line of reflection.

Line: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • MAIN • Q5/5

SET 7 • HSG-CO.A.4

BACKWARD 1

Find the midpoint of a horizontal or vertical segment • Grade 6 • The Number System • 6.NS.C.8

Find the midpoint of the segment from (2, 4) to (8, 4).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q1/5

Find the midpoint of the segment from (1, 1) to (1, 9).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q2/5

On a number line, what number is halfway between −3 and 7?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q3/5

True or false?

The midpoint of the segment from (−4, 0) to (6, 0) is (1, 0).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q4/5

True or false?

The midpoint of a segment is closer to one endpoint than to the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B1-Q5

Grade 6 • The Number System • 6.NS.C.8 • Set 7 • BACKWARD 1 • Q5/5

SET 7 • HSG-CO.A.4

BACKWARD 2

Recognize perpendicular and parallel segments on a grid • Grade 4 • Geometry • 4.G.A.1

True or false?

A horizontal segment and a vertical segment are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q1

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q1/5

True or false?

Any two segments of the same length are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q2

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q2/5

Which pair of segments is parallel?

A. From (0, 0) to (3, 1) and from (0, 2) to (3, 3)

B. From (1, 1) to (2, 3) and from (1, 1) to (3, 2)

C. From (0, 0) to (3, 1) and from (0, 0) to (1, 3)

D. From (0, 0) to (4, 0) and from (0, 0) to (0, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q3

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q3/5

A segment goes from (2, 1) to (2, 6). Describe a segment through (5, 3) that is perpendicular to it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q4

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q4/5

Which describes two lines that are both perpendicular to the x-axis?

A. They are parallel

B. They are perpendicular to each other

C. They meet at the origin

D. They are the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B2-Q5

Grade 4 • Geometry • 4.G.A.1 • Set 7 • BACKWARD 2 • Q5/5

SET 7 • HSG-CO.A.4

BACKWARD 3

Reflect and translate points with coordinate rules • Grade 8 • Geometry • 8.G.A.3

Reflect the point (4, −3) across the x-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q1/5

Translate the point $(-1, 5)$ 6 units right and 2 units down.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q2/5

Reflect the point (2, 7) across the line y = x.

A. (7, 2)

B. (−7, −2)

C. (−2, 7)

D. (2, −7)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q3/5

True or false?

Reflecting the point $(-6, 1)$ across the y-axis gives $(6, 1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q4/5

True or false?

Translating (3, 3) by 2 units left gives (3, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 7 • BACKWARD 3 • Q5/5

SET 7 • HSG-CO.A.4

FORWARD 1

Find lines of reflection and images in slanted, non-axis cases • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A reflection maps P(0, 0) to P′(4, 2). Find an equation of the line of reflection.

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q1/5

A translation maps $A(2, -1)$ to $A'(-3, 4)$. Where does it map $B(0, 6)$?

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q2/5

True or false?

A reflection maps P to P′ and Q to Q′, where P and Q are not on the line of reflection. Then PP′ is parallel to QQ′ (or on the same line).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q3/5

A reflection maps A(1, 3) to A′(3, 1). Is the line of reflection y = x? Use the definition to decide.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q4/5

A reflection maps P(−1, 4) to P′(5, 4). Which point is on the line of reflection?

A. (−1, 0)

B. (5, 0)

C. (2, −3)

D. (4, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 1 • Q5/5

SET 7 • HSG-CO.A.4

FORWARD 2

Prove from the definitions that reflections and translations keep lengths • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Points P and Q are on the same side of line ℓ, and line PQ is not perpendicular to ℓ. A reflection across ℓ maps them to P′ and Q′. Let PP′ meet ℓ at M and QQ′ meet ℓ at N. Prove that PQ = P′Q′.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q1/5

A translation maps P to P′ and Q to Q′, where Q is not on line PP′. Prove that PQ = P′Q′.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q2/5

True or false?

A translation maps P to P′ and Q to Q′. Then PP′Q′Q is always a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q3/5

True or false?

Under a reflection, a point not on the line of reflection and its image are on the same side of the line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q4/5

In a proof that a reflection across ℓ keeps the length of segment PQ, which fact is used first?

A. ℓ is the perpendicular bisector of PP′ and of QQ′

B. ℓ bisects $\angle PQP'$

C. ℓ is parallel to PQ

D. ℓ passes through P

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S7-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 7 • FORWARD 2 • Q5/5

Definition of a rotation

MAIN	Geometry main question	**HSG-CO.A.4**
BACKWARD 1	Describe turns in degrees, clockwise and counterclockwise	**4.MD.C.5.b**
BACKWARD 2	Points on a circle are the same distance from its center	**7.G.B.4**
BACKWARD 3	Rotate points 90°, 180°, and 270° about the origin	**8.G.A.3**
FORWARD 1	Rotate about a center that is not the origin	**HSG-CO.A.4**
FORWARD 2	Find the center and angle of a rotation from points and their imag..	**HSG-CO.A.4**

SET 8 • HSG-CO.A.4

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Complete a precise definition. A counterclockwise rotation about point O through angle θ maps each point P ≠ O to the point P′ such that _____ and _____. Point O maps to _____.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q1/5

A rotation about point O maps point P to P′ (P ≠ P′). Which must be true?

A. O is the midpoint of PP′

B. O is on the perpendicular bisector of PP′

C. OP′ = 2 · OP

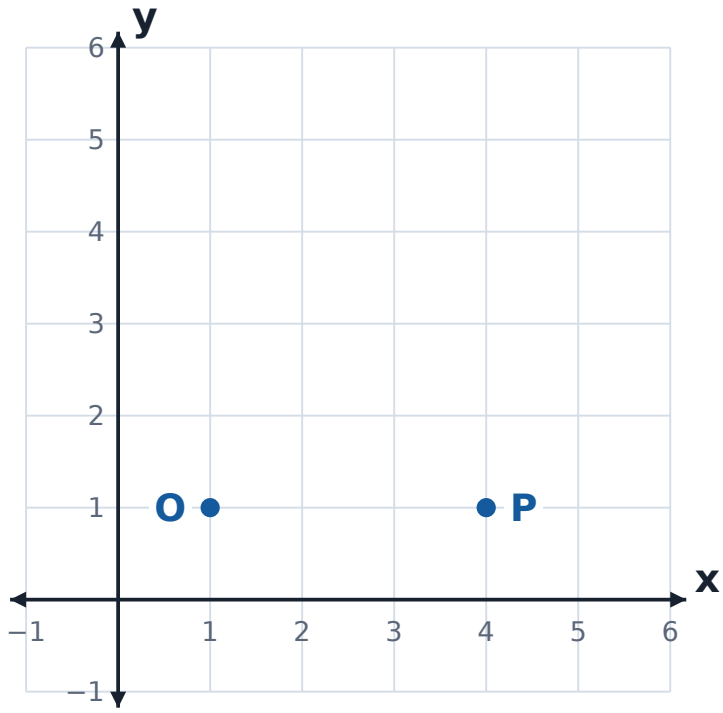
D. PP′ is perpendicular to OP

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q2/5

A rotation of 90° counterclockwise about O(1, 1) maps P(4, 1) to P′. Use the definition to find P′.



P′ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q3/5

True or false?

Under a rotation about point O, each point P moves along the circle with center O that passes through P.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q4/5

A rotation maps P(2, 0) to P′(0, 2). Explain why its center must lie on the line y = x.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • MAIN • Q5/5

SET 8 • HSG-CO.A.4

BACKWARD 1

Describe turns in degrees, clockwise and counterclockwise • Grade 4 • Measurement & Data • 4.MD.C.5.b

A ray points right. It turns 90° counterclockwise. Which way does it point now?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q1

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q1/5

A ray points up. It turns 180°. Which way does it point now?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q2

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q2/5

True or false?

A clockwise turn of 270° ends in the same direction as a counterclockwise turn of 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q3

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q3/5

A ray points up. It turns 90° clockwise. Which way does it point now?

A. Down

B. Left

C. Right

D. Up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q4

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q4/5

How many degrees are in $\frac{3}{4}$ of a full turn?

Degrees: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B1-Q5

Grade 4 • Measurement & Data • 4.MD.C.5.b • Set 8 • BACKWARD 1 • Q5/5

SET 8 • HSG-CO.A.4

BACKWARD 2

Points on a circle are the same distance from its center • Grade 7 • Geometry • 7.G.B.4

A circle has center O and radius 6. Point P is on the circle. What is OP?

OP = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q1/5

True or false?

If OA = OB, then A and B lie on one circle with center O.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q2/5

Which point is on the circle with center (0, 0) and radius 5?

A. (4, 4)

B. (3, 4)

C. (2, 3)

D. (5, 5)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q3/5

A circle has center O and diameter 10 cm. How far is every point of the circle from O?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q4/5

True or false?

Two points the same distance from O are always the same point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 8 • BACKWARD 2 • Q5/5

SET 8 • HSG-CO.A.4

BACKWARD 3

Rotate points 90°, 180°, and 270° about the origin • Grade 8 • Geometry • 8.G.A.3

Rotate (5, 2) by 90° counterclockwise about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q1/5

Rotate $(-3, 4)$ by $180°$ about the origin.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q2/5

Rotate (1, 6) by 90° clockwise about the origin.

A. (−6, 1)

B. (1, −6)

C. (−1, −6)

D. (6, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q3/5

True or false?

Rotating (2, 3) by 270° counterclockwise about the origin gives (3, −2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q4/5

True or false?

Rotating (4, 0) by 90° counterclockwise about the origin gives (0, −4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 8 • BACKWARD 3 • Q5/5

SET 8 • HSG-CO.A.4

FORWARD 1

Rotate about a center that is not the origin • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

Rotate P(5, 4) by 90° clockwise about C(2, 3).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q1/5

Rotate P(1, 5) by 180° about C(−1, 2).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q2/5

Rotate P(3, −2) by 90° counterclockwise about C(0, −2).

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q3/5

Rotate P(1, 4) by 270° counterclockwise about C(1, 1).

A. (4, 4)

B. (4, 1)

C. (1, −2)

D. (−2, 1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q4/5

True or false?

Rotating P(4, 0) by 90° counterclockwise about C(2, 0) gives the same image as rotating P by 90° counterclockwise about the origin.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 1 • Q5/5

SET 8 • HSG-CO.A.4

FORWARD 2

Find the center and angle of a rotation from points and their images • High School — Geometry • Geometry: Congruence • HSG-CO.A.4

A rotation maps B(1, 0) to B′(0, 1) and A(3, 1) to A′(−1, 3). The center is where the perpendicular bisectors of BB′ and AA′ meet. Find the center and the angle of rotation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q1/5

A rotation maps P(4, 1) to P′(2, 3) and Q(4, 3) to Q′(0, 3). Find the center and the angle of rotation.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q2/5

True or false?

The center of a nonzero rotation moves to a new position under the rotation.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q3/5

A rotation about C maps P to P′. $CP = 5$ and $PP' = 5\sqrt{2}$. Which could be the angle of rotation?

A. 60°

B. 90°

C. 45°

D. 180°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q4/5

A rotation about the origin maps (5, 0) to (3, 4). What is the counterclockwise angle of rotation, to the nearest degree?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S8-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.4 • Set 8 • FORWARD 2 • Q5/5

Draw the image of a figure under a rotation, reflection, or translation

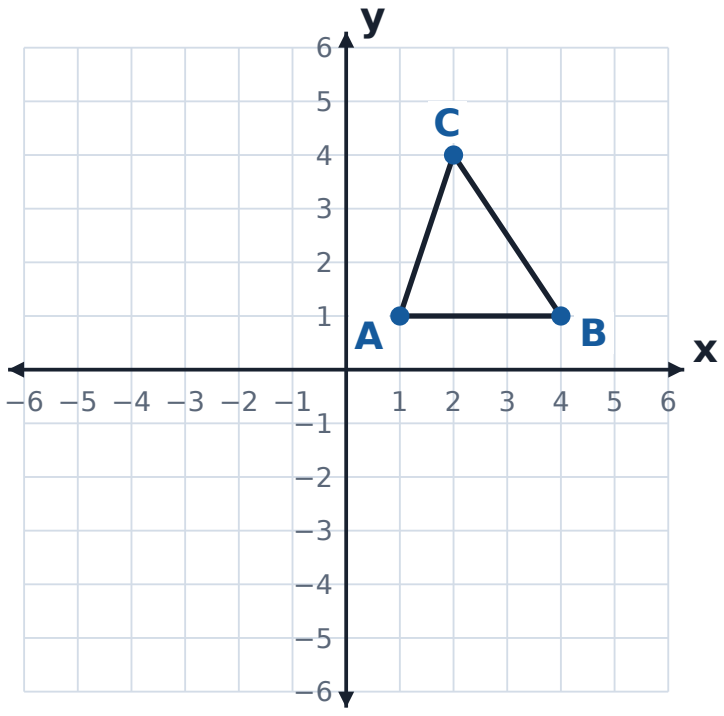
MAIN	Geometry main question	**HSG-CO.A.5**
BACKWARD 1	Plot and read points in all four quadrants	**6.NS.C.6.c**
BACKWARD 2	Find image points with coordinate rules	**8.G.A.3**
BACKWARD 3	Count the distance from a point to a horizontal or vertical line	**6.NS.C.8**
FORWARD 1	Draw the image under a sequence of two transformations	**HSG-CO.A.5**
FORWARD 2	Draw rotations and reflections off the grid with compass, protract…	**HSG-CO.A.5**

SET 9 • HSG-CO.A.5

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Draw the image of triangle ABC under the translation 5 units left and 2 units down. Label the image A′B′C′.

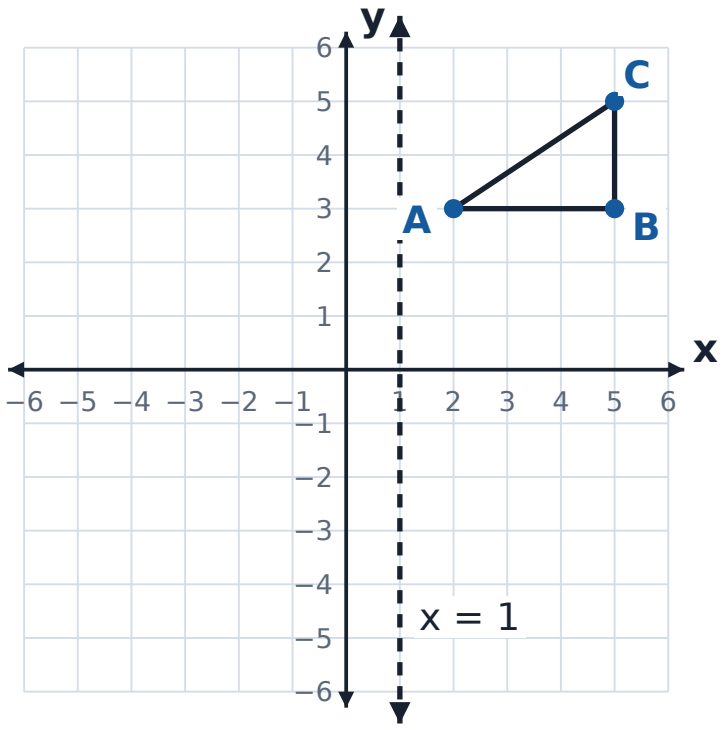


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q1/5

Draw the image of triangle ABC under the reflection across the line x = 1. Label the image A′B′C′.

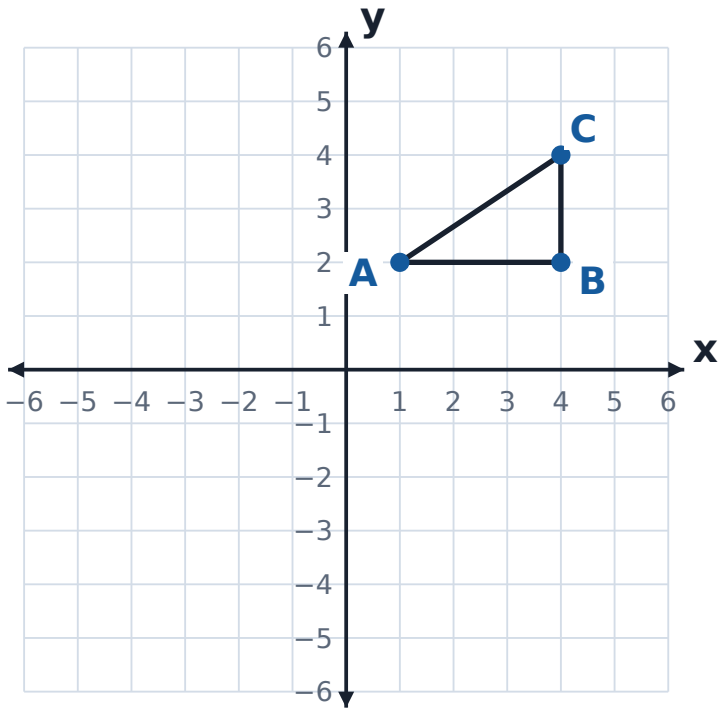


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q2/5

Draw the image of triangle ABC under the rotation of 90° counterclockwise about the origin. Label the image A′B′C′.

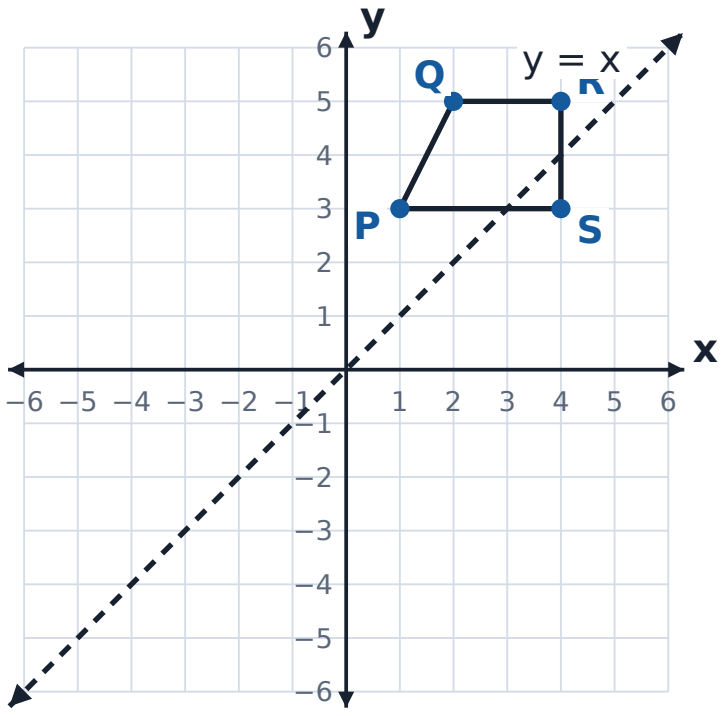


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q3/5

Draw the image of quadrilateral PQRS under the reflection across the line $y = x$. Label the image $P'Q'R'S'$.

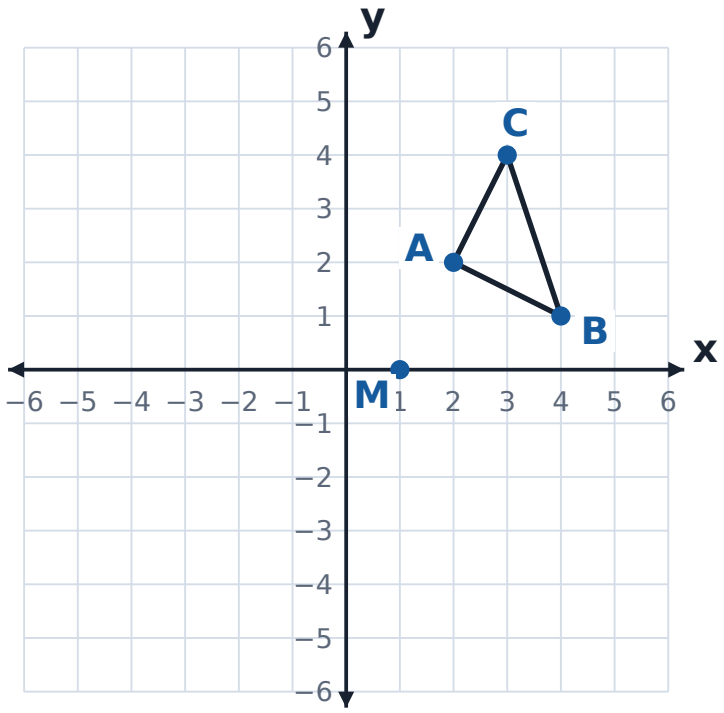


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q4/5

Draw the image of triangle ABC under the rotation of 180° about the point M(1, 0). Label the image A′B′C′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-M-Q5

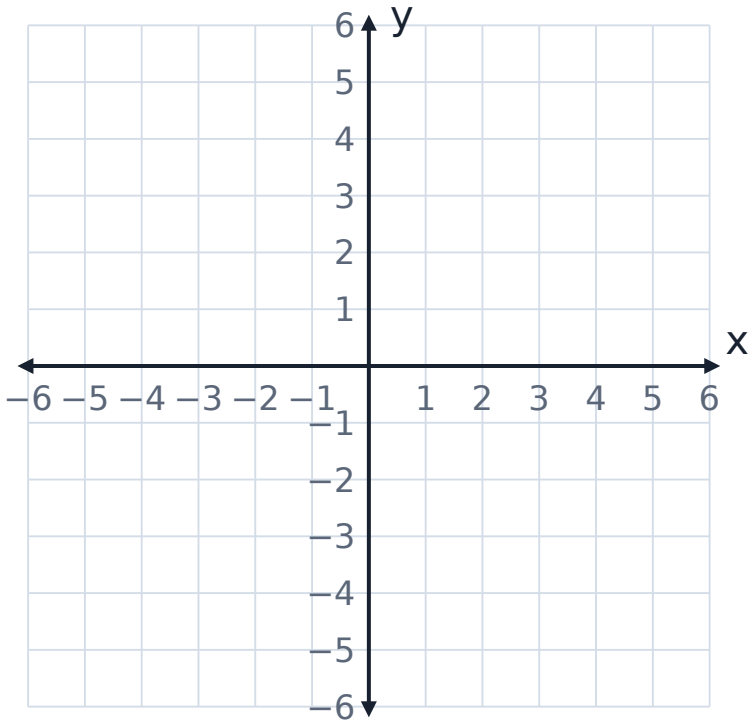
High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • MAIN • Q5/5

SET 9 • HSG-CO.A.5

BACKWARD 1

Plot and read points in all four quadrants • Grade 6 • The Number System • 6.NS.C.6.c

Plot and label the points A(−3, 2), B(0, −4), and C(5, 1).



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q1

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q1/5

A point is 4 units left of the y-axis and 1 unit below the x-axis. What are its coordinates?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q2

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q2/5

Which point is 2 units above the x-axis and on the y-axis?

A. (2, 2)

B. (0, −2)

C. (0, 2)

D. (2, 0)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q3

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q3/5

True or false?

The point (3, −2) is in Quadrant IV.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q4

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q4/5

True or false?

To plot (2, 5), start at the origin and move 2 units up and 5 units right.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B1-Q5

Grade 6 • The Number System • 6.NS.C.6.c • Set 9 • BACKWARD 1 • Q5/5

SET 9 • HSG-CO.A.5

BACKWARD 2

Find image points with coordinate rules • Grade 8 • Geometry • 8.G.A.3

Find the image of $(3, -2)$ under a translation 4 units left and 1 unit up.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q1/5

Find the image of $(-2, 5)$ under a reflection across the x-axis.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q2/5

Find the image of (3, 1) under a 90° clockwise rotation about the origin.

A. (−3, −1)

B. (3, −1)

C. (1, −3)

D. (−1, 3)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q3/5

True or false?

The image of $(4, -1)$ under a reflection across the y-axis is $(-4, -1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q4/5

True or false?

The image of (2, 2) under a 180° rotation about the origin is (2, −2).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 9 • BACKWARD 2 • Q5/5

SET 9 • HSG-CO.A.5

BACKWARD 3

Count the distance from a point to a horizontal or vertical line • Grade 6 • The Number System • 6.NS.C.8

How far is the point (5, 3) from the vertical line x = 1?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q1

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q1/5

How far is the point $(-2, -1)$ from the horizontal line $y = 3$?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q2

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q2/5

Which point is 3 units from the line x = 2, on its left?

A. (−3, 4)

B. (−1, 4)

C. (2, 1)

D. (5, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q3

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q3/5

True or false?

The point (0, 4) is 4 units from the x-axis.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q4

Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q4/5

True or false?

The point (6, 2) is 6 units from the line $x = 2$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-B3-Q5

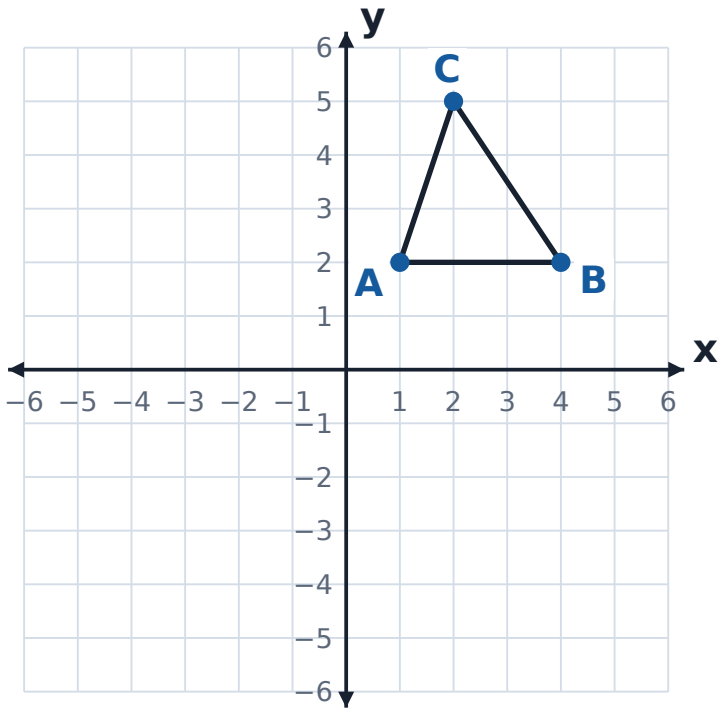
Grade 6 • The Number System • 6.NS.C.8 • Set 9 • BACKWARD 3 • Q5/5

SET 9 • HSG-CO.A.5

FORWARD 1

Draw the image under a sequence of two transformations • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Reflect triangle ABC across the y-axis, then translate the result 3 units down. Draw and label the final image A″B″C″.

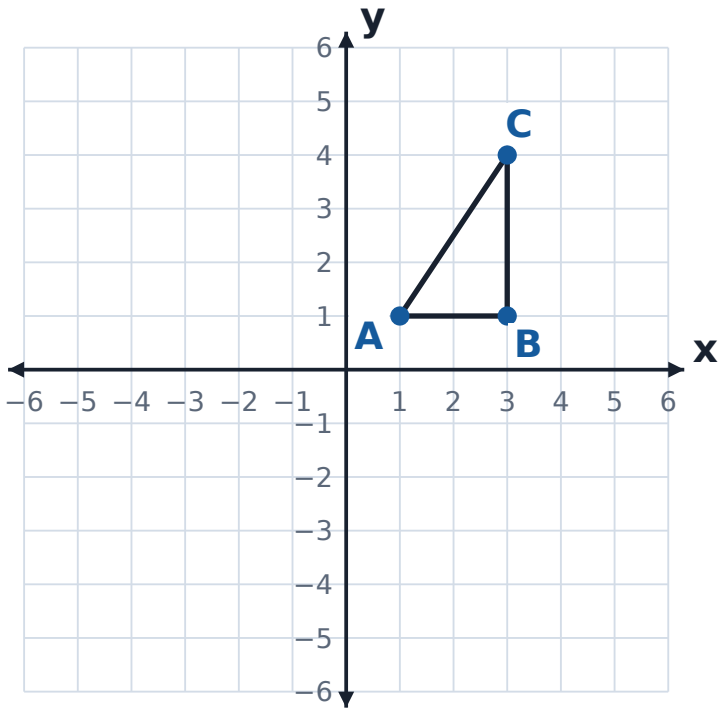


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q1/5

Rotate triangle ABC by 90° counterclockwise about the origin, then reflect the result across the x-axis. Draw and label the final image A″B″C″.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q2/5

Triangle ABC has A(2, 1). It is translated 2 units right and then rotated 180° about the origin. Where does A end up?

A″ = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q3/5

True or false?

Reflecting a figure across the x-axis and then across the y-axis gives the same image as reflecting it across the line y = x.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q4/5

Point P(1, 3) is translated 2 units left and then reflected across the line y = x. Where does it end up?

A. (3, 1)

B. (−1, 3)

C. (1, −3)

D. (3, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 1 • Q5/5

SET 9 • HSG-CO.A.5

FORWARD 2

Draw rotations and reflections off the grid with compass, protractor, and straightedge • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Use a compass and protractor. Rotate point P 60° counterclockwise about point O. Mark and label the image P′.

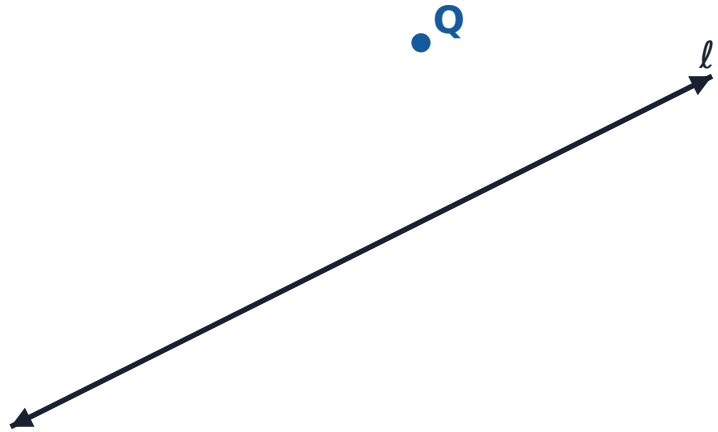


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q1/5

Use a compass and straightedge. Reflect point Q across line ℓ. Mark and label the image Q′.

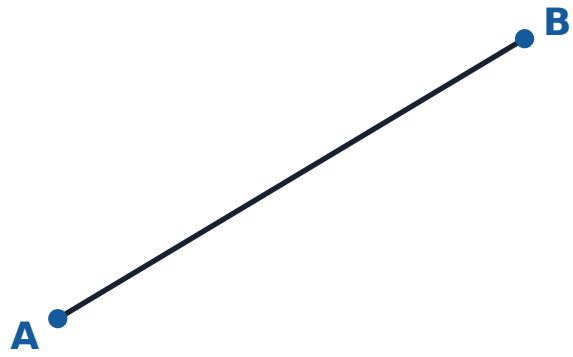


REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q2/5

Use a compass and protractor. Rotate segment AB 90° clockwise about point A. Draw and label the image AB′.



REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q3/5

True or false?

To rotate point P about point O by 45°, it is enough to draw a 45° angle at O; the distance from O does not matter.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q4/5

Describe how to reflect point P across a slanted line ℓ using only a compass and straightedge.

Steps: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S9-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 9 • FORWARD 2 • Q5/5

Specify a sequence of transformations that carries one figure onto another

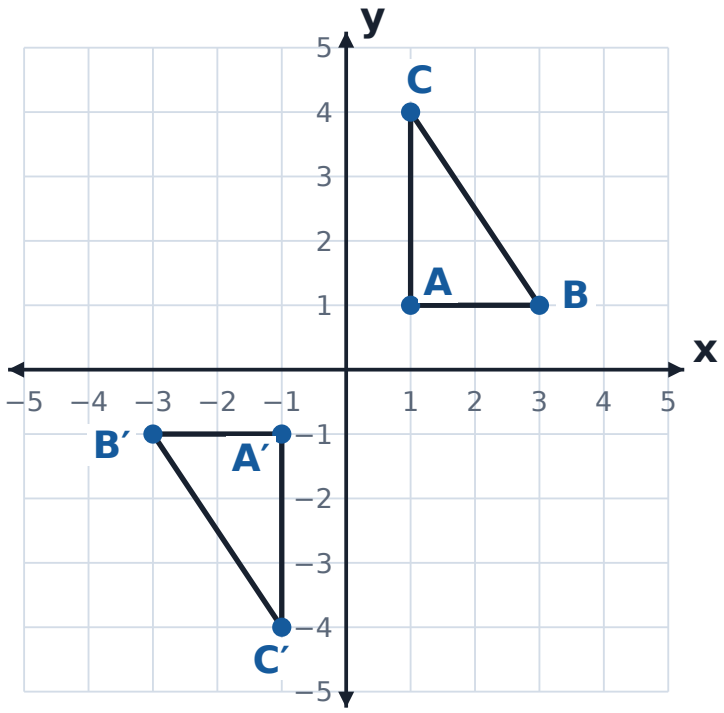
MAIN	Geometry main question	**HSG-CO.A.5**
BACKWARD 1	Understand congruence as the result of one rigid motion	**8.G.A.2**
BACKWARD 2	Find images with coordinate rules for sequences	**8.G.A.3**
BACKWARD 3	Read the orientation (clockwise or counterclockwise order) of a tr…	**8.G.A.1**
FORWARD 1	Use the fewest transformations and justify	**HSG-CO.A.5**
FORWARD 2	Specify a similarity sequence: a dilation and rigid motions	**HSG-SRT.A.2**

SET 10 • HSG-CO.A.5

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Describe a sequence of rigid transformations that carries triangle ABC onto triangle A′B′C′ (A to A′, B to B′, C to C′).



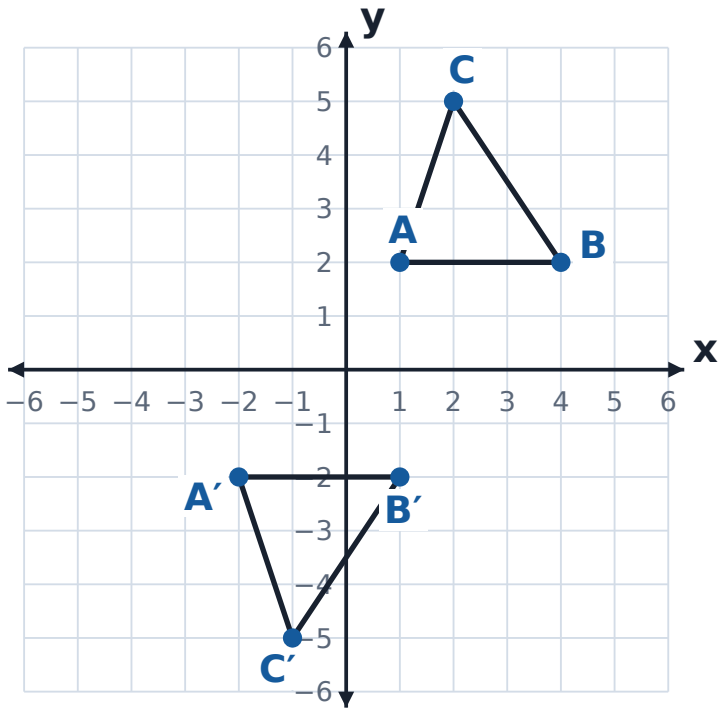
Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q1/5

Describe a sequence of rigid transformations that carries triangle ABC onto triangle A′B′C′.



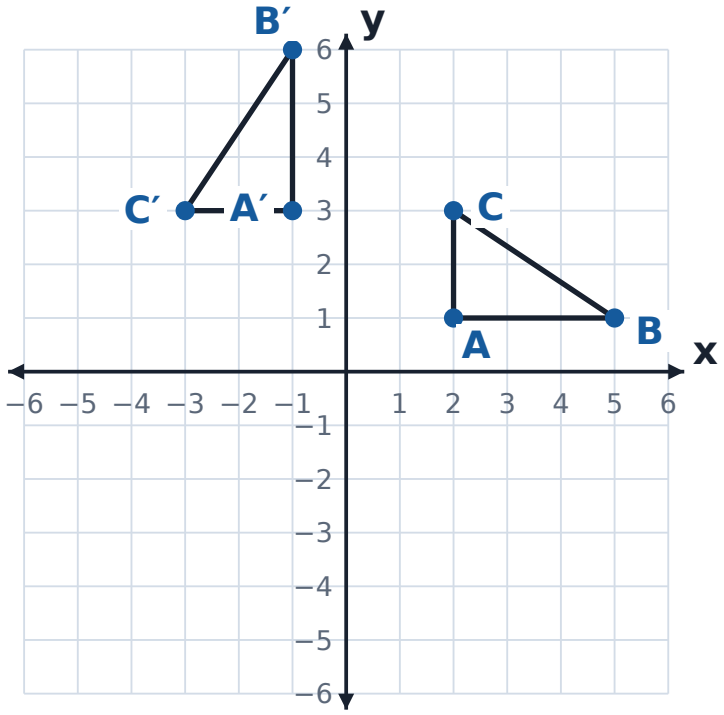
Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q2/5

On the grid, triangle A′B′C′ is congruent to triangle ABC. Describe a sequence of rigid transformations that carries ABC onto A′B′C′, with A → A′, B → B′, C → C′.



Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q3/5

Right triangle ABC has A(0, 0), B(2, 0), C(0, 3). Triangle A′B′C′ has A′(4, 0), B′(4, 2), C′(1, 0). Which sequence carries ABC onto A′B′C′?

A. Reflect across the y-axis, then translate 4 units right

B. Rotate 90° counterclockwise about the origin, then translate 4 units right

C. Rotate 90° clockwise about the origin, then translate 4 units right

D. Translate 4 units right, then rotate 90° counterclockwise about the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q4/5

True or false?

Triangles PQR and P′Q′R′ are congruent. P, Q, R are in clockwise order and P′, Q′, R′ are in counterclockwise order. Some sequence of rotations and translations alone carries P to P′, Q to Q′, and R to R′.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • MAIN • Q5/5

SET 10 • HSG-CO.A.5

BACKWARD 1

Understand congruence as the result of one rigid motion • Grade 8 • Geometry • 8.G.A.2

Triangle 2 is the image of triangle 1 after one rigid motion. Every point moved 3 units right. Which motion was it?

A. A reflection

B. A translation

C. A rotation

D. A dilation

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q1/5

A triangle and its image face opposite ways, as in a mirror. Which kind of rigid motion was most likely used?

Motion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q2/5

True or false?

If one figure is the image of another under a sequence of rigid motions, the figures are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q3/5

True or false?

A figure and its image under a rigid motion can have different areas.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q4/5

Segment AB goes from (0, 0) to (2, 0). Its image goes from (0, 0) to (0, 2). Which single rigid motion could do this?

A. A translation 2 units up

B. A reflection across the x-axis

C. A 90° counterclockwise rotation about the origin

D. A 180° rotation about the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 10 • BACKWARD 1 • Q5/5

SET 10 • HSG-CO.A.5

BACKWARD 2

Find images with coordinate rules for sequences • Grade 8 • Geometry • 8.G.A.3

Point (2, −3) is reflected across the y-axis. Find its image.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q1/5

Point $(-1, 4)$ is rotated 90° counterclockwise about the origin. Find its image.

Image: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q2/5

Point (5, 1) is translated 2 units left and 3 units down. Find its image.

A. (3, −2)

B. (7, 4)

C. (7, −2)

D. (3, 4)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q3/5

True or false?

Point (0, 3) rotated 180° about the origin is (0, −3).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q4/5

True or false?

Point (4, 1) reflected across the line $y = x$ is $(4, -1)$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B2-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 10 • BACKWARD 2 • Q5/5

SET 10 • HSG-CO.A.5

BACKWARD 3

Read the orientation (clockwise or counterclockwise order) of a triangle • Grade 8 • Geometry • 8.G.A.1

Triangle ABC has A(0, 0), B(4, 0), C(0, 3). Going A → B → C, is the order clockwise or counterclockwise?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q1/5

True or false?

A reflection reverses the orientation of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q2/5

True or false?

A rotation reverses the orientation of a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q3/5

Triangle PQR is clockwise. After two reflections, its image is

A. counterclockwise

B. either, depending on the lines

C. clockwise

D. not a triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q4/5

Triangle ABC is counterclockwise. It is translated and then reflected. Is the final image clockwise or counterclockwise?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 10 • BACKWARD 3 • Q5/5

SET 10 • HSG-CO.A.5

FORWARD 1

Use the fewest transformations and justify • High School — Geometry • Geometry: Congruence • HSG-CO.A.5

Two reflections are done one after the other, across parallel lines that are 3 units apart. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q1/5

Two reflections are done one after the other, across lines that meet at point O and form a 40° angle. Which single transformation has the same result?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q2/5

Triangle ABC is clockwise and triangle A′B′C′ is congruent to it and counterclockwise. Explain why no single rotation or translation carries ABC onto A′B′C′.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q3/5

True or false?

Every rigid motion of the plane can be written as a sequence of at most three reflections.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q4/5

Triangle A′B′C′ is the image of triangle ABC under some rigid motion, and both are counterclockwise. What is the fewest number of reflections that can always carry ABC onto A′B′C′?

A. 2

B. 4

C. 3

D. 1

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.5 • Set 10 • FORWARD 1 • Q5/5

SET 10 • HSG-CO.A.5

FORWARD 2

Specify a similarity sequence: a dilation and rigid motions • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

Triangle ABC has A(0, 0), B(2, 0), C(0, 1). Triangle A′B′C′ has A′(0, 0), B′(0, 6), C′(−3, 0). Describe a sequence (a dilation and a rigid motion) that carries ABC onto A′B′C′.

Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q1/5

Square S has vertices (0, 0), (1, 0), (1, 1), (0, 1). Square T has vertices (2, 2), (4, 2), (4, 4), (2, 4). Describe a similarity sequence carrying S onto T.

Sequence: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q2/5

True or false?

Any two figures that a sequence of rigid motions and a dilation carries one onto the other are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q3/5

Segment AB is from (0, 0) to (3, 0). Segment CD is from (1, 1) to (1, 7). Which sequence carries AB onto CD (A to C, B to D)?

A. Rotate 90° counterclockwise about the origin, then translate 1 right and 1 up

B. Dilate by 2 about the origin, then translate 1 right and 1 up

C. Dilate by 3 about the origin, rotate 90° counterclockwise, then translate 1 right and 1 up

D. Dilate by 2 about the origin, rotate 90° counterclockwise about the origin, then translate 1 right and 1 up

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q4/5

Triangle PQR has sides 3, 4, 5. Triangle XYZ has sides 6, 8, 10. What scale factor must the dilation in a similarity sequence from PQR to XYZ have?

Scale factor: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S10-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 10 • FORWARD 2 • Q5/5

Predict the effect of a rigid motion on a figure

MAIN	Geometry main question	**HSG-CO.B.6**
BACKWARD 1	Know what rigid motions keep: lines, lengths, angles, and parallel l…	**8.G.A.1**
BACKWARD 2	Match the vertices and sides of a figure with its image	**8.G.A.2**
FORWARD 1	Predict the effect of a sequence of rigid motions	**HSG-CO.B.6**
FORWARD 2	Use a rigid motion to conclude that parts of a figure are congruent	**HSG-CO.B.7**

SET 11 • HSG-CO.B.6

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Triangle ABC has AB = 7, BC = 5, and m∠B = 48°. It is reflected across a line ℓ. What are A′B′ and m∠B′?

A. 14 and 48°

B. 7 and 132°

C. 7 and 48°

D. They depend on where ℓ is

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q1/5

Quadrilateral PQRS has PQ ∥ SR. It is rotated 90° about a point O. What is true about P′Q′ and S′R′? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q2/5

Point A is 4 cm from line m. Triangle ABC is reflected across m. How far is A′ from m, and on which side of m is it?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q3/5

True or false?

A translation can reverse the orientation (the clockwise or counterclockwise order) of a triangle's vertices.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q4/5

Segment AB lies on line ℓ. A 180° rotation about a point O on ℓ maps AB to A′B′. Where does A′B′ lie? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • MAIN • Q5/5

SET 11 • HSG-CO.B.6

BACKWARD 1

Know what rigid motions keep: lines, lengths, angles, and parallel lines • Grade 8 • Geometry • 8.G.A.1

True or false?

Under a rotation, the image of a line is a line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q1/5

True or false?

A reflection can turn two parallel lines into two lines that meet.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q2/5

A 52° angle is translated 6 units to the right. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q3/5

A 9 cm segment is rotated 30° about one of its endpoints. How long is its image?

A. 30 cm

B. 4.5 cm

C. 39 cm

D. 9 cm

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q4/5

Name two properties of a figure that a reflection does NOT change.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B1-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 11 • BACKWARD 1 • Q5/5

SET 11 • HSG-CO.B.6

BACKWARD 2

Match the vertices and sides of a figure with its image • Grade 8 • Geometry • 8.G.A.2

Triangle ABC is rotated to triangle A′B′C′. Which side of the image corresponds to side BC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q1/5

Quadrilateral PQRS is reflected to P′Q′R′S′. Which angle corresponds to ∠Q?

A. ∠R′

B. ∠P′

C. ∠Q′

D. ∠S′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q2/5

True or false?

If triangle ABC is translated to triangle A′B′C′, then AB = A′B′.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q3/5

Triangle DEF is reflected to D′E′F′. DE = 6 and EF = 9. Find D′E′ and E′F′.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q4/5

True or false?

After a rotation, side A′B′ of the image always lies on top of side AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-B2-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 11 • BACKWARD 2 • Q5/5

SET 11 • HSG-CO.B.6

FORWARD 1

Predict the effect of a sequence of rigid motions • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Triangle ABC is reflected across line ℓ and then across line m, where $\ell \parallel$ m. Is the final image clockwise or counterclockwise if ABC is counterclockwise? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q1/5

A figure is rotated 70° about point O and then rotated 50° about the same point O. Which single rigid motion has the same effect?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q2/5

True or false?

A translation that is not the identity has exactly one fixed point.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q3/5

Which rigid motion keeps exactly one point fixed?

A. A translation 3 units right

B. A reflection across line ℓ

C. A rotation of 90° about point O

D. The identity

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q4/5

Line ℓ is reflected across line m, and ℓ is perpendicular to m. What is the image of ℓ? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 11 • FORWARD 1 • Q5/5

SET 11 • HSG-CO.B.6

FORWARD 2

Use a rigid motion to conclude that parts of a figure are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rotation about O maps triangle OAB onto triangle OCD, with A → C and B → D. OA = 5 and AB = 7. Find OC and CD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q1/5

A reflection across line ℓ maps triangle PQR onto triangle PSR (Q → S, and P and R are on ℓ). $m\angle PQR = 64°$. Find $m\angle PSR$.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q2/5

True or false?

If a rigid motion maps segment AB onto segment CD, then AB could be longer than CD.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q3/5

A translation maps triangle KLM onto triangle K′L′M′. Which conclusion is NOT justified?

A. KL ∥ K′L′ or KL is on line K′L′

B. ∠L ≅ ∠L′

C. KL is on the same line as K′L′

D. KL = K′L′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q4/5

Explain why the image of an isosceles triangle under any rigid motion is also isosceles.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S11-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 11 • FORWARD 2 • Q5/5

Decide whether two figures are congruent using rigid motions

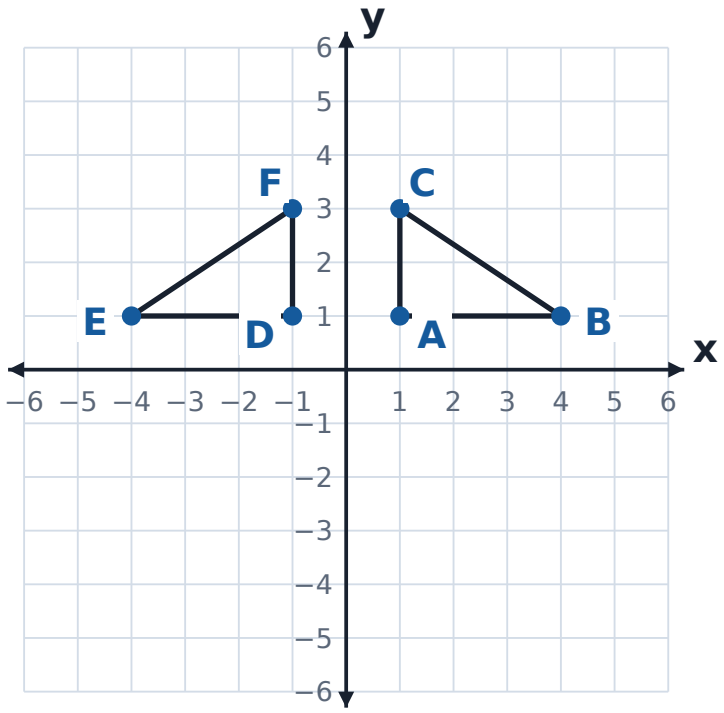
MAIN	Geometry main question	**HSG-CO.B.6**
BACKWARD 1	Recognize congruent figures as images under rigid motions	**8.G.A.2**
BACKWARD 2	Find lengths on a grid with the Pythagorean Theorem	**8.G.B.8**
FORWARD 1	Decide congruence for figures given by measurements, including…	**HSG-CO.B.6**
FORWARD 2	Use an established congruence to find unknown parts	**HSG-CO.B.7**

SET 12 • HSG-CO.B.6

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Is triangle ABC congruent to triangle DEF? If so, give a rigid motion that carries ABC onto DEF. If not, explain why.



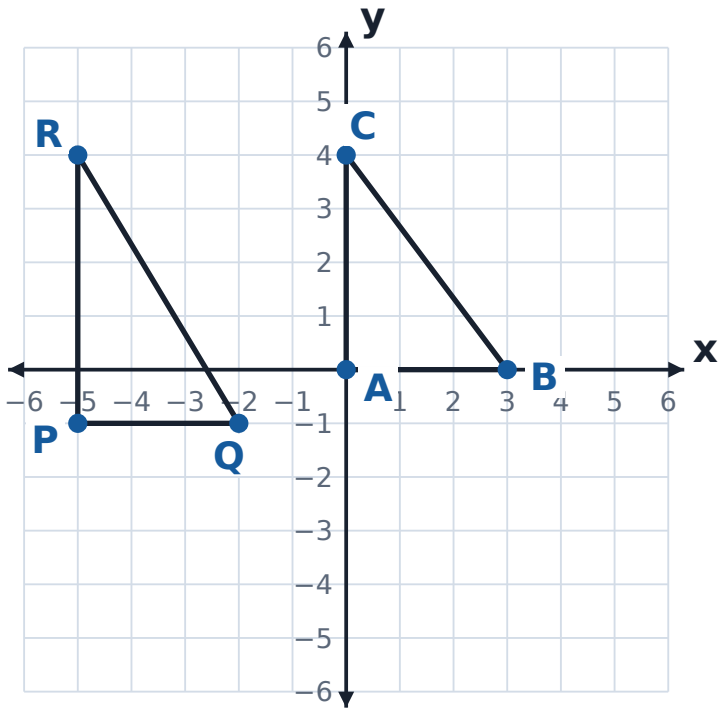
Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q1/5

Is triangle ABC congruent to triangle PQR? If so, give a rigid motion that carries one onto the other. If not, explain why.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q2/5

Which fact proves that two figures are NOT congruent?

A. One is above the other

B. One has area 12 and the other has area 15

C. They face opposite directions

D. They are in different quadrants

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q3/5

True or false?

Two figures are congruent if some sequence of rotations, reflections, and translations carries one onto the other.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q4/5

Rectangle R is 4 cm wide and 6 cm tall. Rectangle S is 6 cm wide and 4 cm tall. Are they congruent? Give a rigid motion or explain why not.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • MAIN • Q5/5

SET 12 • HSG-CO.B.6

BACKWARD 1

Recognize congruent figures as images under rigid motions • Grade 8 • Geometry • 8.G.A.2

True or false?

If one figure is a translation of another, the figures are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q1/5

True or false?

Two figures with the same shape but different sizes are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q2/5

Which pair of figures must be congruent?

A. A square and a larger square

B. A triangle and its reflection

C. A circle of radius 2 and a circle of radius 3

D. A triangle and a dilation of it by 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q3/5

A triangle has sides 5, 6, and 7. It is rotated 90°. What are the side lengths of its image?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q4/5

Is a figure congruent to its image after a reflection followed by a translation?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 12 • BACKWARD 1 • Q5/5

SET 12 • HSG-CO.B.6

BACKWARD 2

Find lengths on a grid with the Pythagorean Theorem • Grade 8 • Geometry • 8.G.B.8

Find the length of the segment from (1, 2) to (5, 2).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q1/5

Find the length of the segment from (0, 0) to (3, 4).

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q2/5

Find the length of the segment from $(-2, 1)$ to $(1, 5)$.

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q3/5

True or false?

The segments from (0, 0) to (2, 1) and from (0, 0) to (1, 2) have the same length.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q4/5

True or false?

The segment from (0, 0) to (3, 3) has length 3.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-B2-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 12 • BACKWARD 2 • Q5/5

SET 12 • HSG-CO.B.6

FORWARD 1

Decide congruence for figures given by measurements, including curved figures • High School — Geometry • Geometry: Congruence • HSG-CO.B.6

Circle P has radius 3 and circle Q has radius 3. Explain, using a rigid motion, why they are congruent.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q1/5

Two quadrilaterals each have sides 5, 5, 5, 5. One is a square and the other has a 60° angle. Are they congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q2/5

True or false?

Two triangles with the same perimeter must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q3/5

Which information is enough to show two segments are congruent?

A. They have the same length

B. They have the same slope

C. They have the same midpoint

D. They both start at the origin

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q4/5

Angle 1 measures 35° and angle 2 measures 35°. Describe a rigid motion that carries angle 1 onto angle 2.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.6 • Set 12 • FORWARD 1 • Q5/5

SET 12 • HSG-CO.B.6

FORWARD 2

Use an established congruence to find unknown parts • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rigid motion carries quadrilateral ABCD onto quadrilateral EFGH with A → E, B → F, C → G, D → H. AB = 6, ∠C = 110°, and GH = 4. Find EF, ∠G, and CD.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q1/5

Triangle RST is congruent to triangle XYZ by a rigid motion with R → X, S → Y, T → Z. RS = 2x + 3 and XY = 15. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q2/5

True or false?

If a rigid motion carries triangle ABC onto triangle DEF with A → D, B → E, C → F, then ∠B ≅ ∠F.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q3/5

A reflection carries pentagon PQRST onto pentagon P′Q′R′S′T′. Which pair must be congruent?

A. ∠S and ∠S′

B. PQ and RS

C. QR and S′T′

D. ∠S and ∠T′

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q4/5

Triangle ABC is congruent to triangle CDA by a rigid motion with A → C, B → D, C → A. m∠BAC = 3y° and m∠DCA = 42°. Find y.

y = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S12-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 12 • FORWARD 2 • Q5/5

Congruent triangles: a rigid motion exactly when all corresponding parts are congruent

MAIN	Geometry main question	**HSG-CO.B.7**
BACKWARD 1	Read a congruence statement to match corresponding parts	**8.G.A.2**
BACKWARD 2	Rigid motions keep lengths and angle measures	**8.G.A.1**
BACKWARD 3	Move a point onto another and a ray onto another with rigid motio…	**8.G.A.3**
FORWARD 1	Find which three corresponding parts are enough	**HSG-CO.B.8**
FORWARD 2	Use congruent triangles to prove a relationship in a figure	**HSG-SRT.B.5**

SET 13 • HSG-CO.B.7

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

A rigid motion carries $\triangle ABC$ onto $\triangle DEF$ with $A \to D$, $B \to E$, $C \to F$. Explain why $AB = DE$, $BC = EF$, $CA = FD$, $\angle A \cong \angle D$, $\angle B \cong \angle E$, and $\angle C \cong \angle F$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q1/5

In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $BC = EF$, $CA = FD$, $\angle A \cong \angle D$, $\angle B \cong \angle E$, and $\angle C \cong \angle F$. Explain how to build a rigid motion that carries $\triangle ABC$ onto $\triangle DEF$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q2/5

A rigid motion maps $\triangle PQR$ onto $\triangle XYZ$ with $P \to X$, $Q \to Y$, $R \to Z$. Which must be true?

A. $QR = YZ$

B. $PQ = YZ$

C. $\angle P \cong \angle Y$

D. $QR = XZ$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q3/5

$\triangle JKL \cong \triangle MNO$ with $J \leftrightarrow M$, $K \leftrightarrow N$, $L \leftrightarrow O$. $JK = 8$, $KL = 11$, and $m\angle L = 37°$. Find MN, NO, and $m\angle O$.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q4/5

True or false?

Two triangles whose three pairs of corresponding angles are congruent must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 13 • MAIN • Q5/5

SET 13 • HSG-CO.B.7

BACKWARD 1

Read a congruence statement to match corresponding parts • Grade 8 • Geometry • 8.G.A.2

△ABC ≅ △XYZ. Which angle corresponds to ∠B?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q1/5

△ABC ≅ △XYZ. Which side corresponds to AC?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q2/5

△PQR ≅ △LMN. Which pair corresponds?

A. QR and MN

B. QR and LN

C. PQ and MN

D. PR and LM

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q3/5

True or false?

$\triangle DEF \cong \triangle GHI$ means that $\angle F \cong \angle I$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q4/5

True or false?

$\triangle DEF \cong \triangle GHI$ means that $DE = HI$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B1-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 13 • BACKWARD 1 • Q5/5

SET 13 • HSG-CO.B.7

BACKWARD 2

Rigid motions keep lengths and angle measures • Grade 8 • Geometry • 8.G.A.1

True or false?

A rotation keeps the measure of every angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q1/5

Segment PQ is 12 units long. It is reflected and then translated. How long is the image?

Length: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q2/5

A 71° angle is reflected across a line. Its image measures

A. 289°

B. 109°

C. 71°

D. 19°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q3/5

True or false?

A translation can change the length of a segment if it moves the segment far enough.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q4/5

Which three basic rigid motions keep lengths and angle measures?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 13 • BACKWARD 2 • Q5/5

SET 13 • HSG-CO.B.7

BACKWARD 3

Move a point onto another and a ray onto another with rigid motions • Grade 8 • Geometry • 8.G.A.3

Which translation moves (2, 3) onto (5, −1)?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q1

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q1/5

A ray starts at the origin and points along the positive x-axis. Which rotation about the origin makes it point along the positive y-axis?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q2

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q2/5

After a rotation, point C is below line DE but should be above it, and D and E are on the line. Which motion moves C across without moving D or E?

A. A rotation about D

B. A translation along DE

C. A rotation about the midpoint of DE

D. A reflection across line DE

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q3

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q3/5

True or false?

Reflecting across a line keeps every point of that line fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q4

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q4/5

True or false?

Translating a triangle can keep exactly one of its vertices fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-B3-Q5

Grade 8 • Geometry • 8.G.A.3 • Set 13 • BACKWARD 3 • Q5/5

SET 13 • HSG-CO.B.7

FORWARD 1

Find which three corresponding parts are enough • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Must two triangles be congruent if all three pairs of corresponding sides are congruent? Use the result of the full six-part test.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q1/5

Give an example of two triangles with two pairs of congruent sides and a pair of congruent angles that are NOT congruent.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q2/5

True or false?

Knowing that all three pairs of corresponding angles are congruent is enough to prove two triangles congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q3/5

Which set of corresponding parts is NOT always enough for congruence?

A. Three sides

B. Two sides and an angle not between them

C. Two sides and the angle between them

D. Two angles and the side between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q4/5

Two triangles have AB = DE, ∠A ≅ ∠D, and ∠B ≅ ∠E. Must ∠C ≅ ∠F? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 13 • FORWARD 1 • Q5/5

SET 13 • HSG-CO.B.7

FORWARD 2

Use congruent triangles to prove a relationship in a figure • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In quadrilateral ABCD, AB = CB and AD = CD. Prove that diagonal BD bisects ∠ABC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q1/5

Segments AC and BD bisect each other at E. Prove that AB = CD.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q2/5

In $\triangle PQR$ and $\triangle STU$, $PQ = ST$, $QR = TU$, and $PR = SU$. The triangles are congruent. What must be true of $\angle R$ and $\angle U$? Why?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q3/5

True or false?

Once two triangles are proved congruent, only the parts used in the proof are known to be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q4/5

To prove that point M is the midpoint of segment XY using congruent triangles, which pair of parts must you show is congruent?

A. ∠M and ∠X

B. ∠X and ∠Y

C. XM and YM

D. XY and XM

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S13-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 13 • FORWARD 2 • Q5/5

Explain SAS congruence with rigid motions

MAIN	Geometry main question	**HSG-CO.B.8**
BACKWARD 1	Identify the included angle and included side	**7.G.A.2**
BACKWARD 2	Rigid motions map rays, lengths, and angles to equal ones	**8.G.A.1**
BACKWARD 3	Choose a translation, rotation, or reflection to move a point or ray…	**8.G.A.2**
FORWARD 1	Prove triangles congruent by SAS inside a larger figure	**HSG-SRT.B.5**
FORWARD 2	Decide whether given information is enough for SAS, and name a…	**HSG-SRT.B.5**

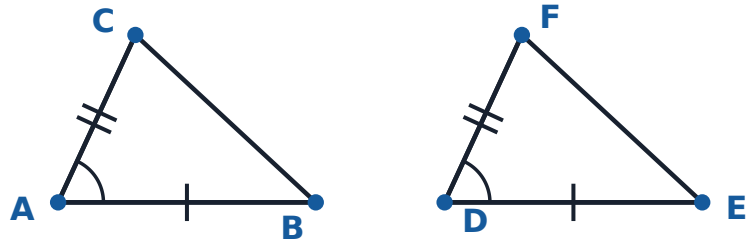
SET 14 • HSG-CO.B.8

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $AC = DF$, and $\angle A \cong \angle D$. Explain, using rigid motions, why $\triangle ABC \cong \triangle DEF$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q1/5

In the rigid-motion proof of SAS, A has been moved to D and ray AB lies on ray DE, with C on the same side of line DE as F. Why must ray AC lie on ray DF?

A. BC = EF

B. The triangles have equal areas

C. AC = DF

D. $\angle A \cong \angle D$, so both rays make the same angle with ray DE on the same side

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q2/5

In the rigid-motion proof of SAS, ray AC now lies on ray DF. Why does C land exactly on F?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q3/5

Explain why a reflection is sometimes needed in the rigid-motion proof of SAS, and why it does not move D or E.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q4/5

True or false?

The rigid-motion argument for SAS still works if the congruent angle is not between the two congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 14 • MAIN • Q5/5

SET 14 • HSG-CO.B.8

BACKWARD 1

Identify the included angle and included side • Grade 7 • Geometry • 7.G.A.2

In △ABC, which angle is included between sides AB and AC?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q1/5

In △PQR, which side is included between ∠P and ∠Q?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q2/5

In △XYZ, which angle is NOT included between sides XY and YZ?

A. Both ∠X and ∠Y

B. ∠X

C. ∠Y

D. None of the angles is excluded

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q3/5

True or false?

Two sides of 5 cm and 7 cm with a 40° angle between them determine one triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q4/5

True or false?

Two sides of 5 cm and 7 cm with a 40° angle that is not between them always determine one triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 14 • BACKWARD 1 • Q5/5

SET 14 • HSG-CO.B.8

BACKWARD 2

Rigid motions map rays, lengths, and angles to equal ones • Grade 8 • Geometry • 8.G.A.1

True or false?

Under a rigid motion, the image of a ray is a ray.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q1/5

A 30° angle is rotated about its vertex. What is the measure of the image angle?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q2/5

A point is 5 units from D along a ray. After a rotation about D, the image point is

A. closer to D

B. not on the image ray

C. 5 units from D on the image ray

D. farther from D

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q3/5

True or false?

A reflection across line DE can move point D.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q4/5

Segment AB is translated so that A lands on D. AB = 9. How far is the image of B from D?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B2-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 14 • BACKWARD 2 • Q5/5

SET 14 • HSG-CO.B.8

BACKWARD 3

Choose a translation, rotation, or reflection to move a point or ray onto a target • Grade 8 • Geometry • 8.G.A.2

Which kind of rigid motion moves point A onto point D without turning anything?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q1

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q1/5

A and D are already together. Which kind of rigid motion turns ray AB onto ray DE without moving D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q2

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q2/5

D and E are fixed, and point C is on the wrong side of line DE. Which motion fixes D and E and moves C across the line?

A. A translation along DE

B. A rotation of 90° about D

C. A rotation of 180° about E

D. A reflection across line DE

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q3

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q3/5

True or false?

A rotation about point D keeps D fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q4

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q4/5

True or false?

A translation that moves A onto D also keeps A fixed.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-B3-Q5

Grade 8 • Geometry • 8.G.A.2 • Set 14 • BACKWARD 3 • Q5/5

SET 14 • HSG-CO.B.8

FORWARD 1

Prove triangles congruent by SAS inside a larger figure • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In quadrilateral ABCD, AB ∥ DC and AB = DC. Diagonal AC is drawn. Prove that △ABC ≅ △CDA.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q1/5

Point M is the midpoint of segment AB, and line ℓ is perpendicular to AB at M. Point P is on ℓ. Prove that $\triangle PMA \cong \triangle PMB$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q2/5

Segments AC and BD cross at E, with AE = CE and BE = DE. Name the congruent triangles and the criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q3/5

True or false?

In an SAS proof, a side that the two triangles share can be one of the pairs of congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q4/5

Segments PQ and RS bisect each other at T. Which reason gives the included angles for SAS?

A. The angles are both right angles

B. Vertical angles are congruent

C. Base angles are congruent

D. Corresponding angles are congruent

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION **ID S14-F1-Q5**

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 1 • Q5/5

SET 14 • HSG-CO.B.8

FORWARD 2

Decide whether given information is enough for SAS, and name a missing condition • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

In △ABC and △DEF, AB = DE and BC = EF. Name ONE more pair of congruent parts that would prove congruence by SAS.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q1/5

In $\triangle PQR$ and $\triangle XYZ$, $PQ = XY$, $PR = XZ$, and $\angle Q \cong \angle Y$. Is this enough to prove the triangles congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q2/5

True or false?

In $\triangle ABC$ and $\triangle ADC$ (sharing AC), $AB = AD$ and $\angle BCA \cong \angle DCA$. These facts prove $\triangle ABC \cong \triangle ADC$ by SAS.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q3/5

Kites ABCD and EFGH have AB = EF, BC = FG, and ∠B ≅ ∠F. Which triangles are congruent by SAS?

A. △BCD and △FGH

B. △ABC and △EFG

C. △ABD and △EFH

D. △ACD and △EGH

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q4/5

Two triangles have sides 7 and 9 with a 50° angle between them. A student says they need not be congruent because the third sides were not measured. Is the student right? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S14-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 14 • FORWARD 2 • Q5/5

Explain ASA congruence with rigid motions

MAIN	Geometry main question	**HSG-CO.B.8**
BACKWARD 1	Two angles and the included side determine a triangle	**7.G.A.2**
BACKWARD 2	Two different lines meet in at most one point	**8.EE.C.8.a**
BACKWARD 3	Rigid motions keep rays and angle measures	**8.G.A.1**
FORWARD 1	Explain AAS from ASA and the triangle angle sum	**HSG-CO.C.10**
FORWARD 2	Prove congruence with ASA or AAS in overlapping triangles	**HSG-SRT.B.5**

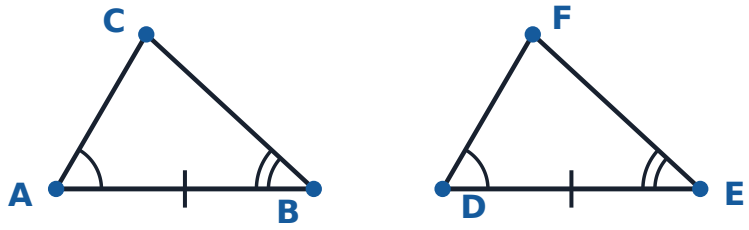
SET 15 • HSG-CO.B.8

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In $\triangle ABC$ and $\triangle DEF$, $\angle A \cong \angle D$, $AB = DE$, and $\angle B \cong \angle E$. Explain, using rigid motions, why $\triangle ABC \cong \triangle DEF$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q1/5

In the rigid-motion proof of ASA, A is on D, B is on E, and C is on F's side of line DE. Why does C land on F?

A. C and F are both the highest points

B. BC = EF

C. C is where rays AC and BC meet; those rays now lie on rays DF and EF, which meet only at F

D. AC = DF

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q2/5

In the rigid-motion proof of ASA, A is on D, B is on E, and C is on the same side of line DE as F. Explain why ray BC lies on ray EF.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q3/5

In the rigid-motion proof of ASA, why can rays DF and EF meet in only one point?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q4/5

True or false?

In the rigid-motion proof of ASA, a reflection across line DE leaves D and E where they are.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 15 • MAIN • Q5/5

SET 15 • HSG-CO.B.8

BACKWARD 1

Two angles and the included side determine a triangle • Grade 7 • Geometry • 7.G.A.2

True or false?

Two angles of 40° and 60° with a 5 cm side between them determine exactly one triangle (up to position).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q1/5

True or false?

Any two angles with a side between them make a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q2/5

A triangle has angles of 50° and 70° at the ends of a 6 cm side. What is the third angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q3/5

Which set of conditions determines exactly one triangle?

A. Angles 30° and 80° with a 4 cm side between them

B. One angle of 45° and nothing else

C. Angles 30°, 60°, and 90°

D. Angles 100° and 90° with a 3 cm side between them

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q4/5

Two angles of a triangle are 95° and 90°. Can the triangle exist? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 15 • BACKWARD 1 • Q5/5

SET 15 • HSG-CO.B.8

BACKWARD 2

Two different lines meet in at most one point • Grade 8 • Expressions & Equations • 8.EE.C.8.a

How many points do the lines $y = x + 1$ and $y = -x + 3$ have in common?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q1/5

True or false?

Two different lines can meet in exactly two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q2/5

Lines $y = 2x + 1$ and $y = 2x - 4$

A. never meet

B. are the same line

C. meet at one point

D. meet at two points

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q3/5

True or false?

If two lines share two points, they are the same line.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q4/5

Where do the lines $x = 2$ and $y = -3$ meet?

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.C.8.a • Set 15 • BACKWARD 2 • Q5/5

SET 15 • HSG-CO.B.8

BACKWARD 3

Rigid motions keep rays and angle measures • Grade 8 • Geometry • 8.G.A.1

A 64° angle is translated and then reflected. What is the measure of its image?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q1/5

True or false?

A rigid motion maps an angle to an angle with the same measure.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q2/5

A ray from E makes a 35° angle with ray ED above line DE. How many such rays are there?

A. Exactly one

B. Two

C. Infinitely many

D. None

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q3/5

True or false?

After a rotation, an angle can become larger if its sides are longer.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q4/5

Segment AB = 4 cm is rotated so that A lands on D. Where is B, relative to D?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 15 • BACKWARD 3 • Q5/5

SET 15 • HSG-CO.B.8

FORWARD 1

Explain AAS from ASA and the triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In $\triangle ABC$ and $\triangle DEF$, $\angle A \cong \angle D$, $\angle B \cong \angle E$, and $BC = EF$ (the side is NOT between the angles). Prove $\triangle ABC \cong \triangle DEF$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q1/5

Two triangles have angles 40° and 75° and a 9 cm side opposite the 75° angle. Must they be congruent? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q2/5

True or false?

AAS is a valid congruence criterion only because the angles of a triangle add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q3/5

In $\triangle PQR$ and $\triangle XYZ$, $\angle P \cong \angle X$, $\angle Q \cong \angle Y$, and $PR = XZ$. Which criterion proves congruence?

A. ASA

B. AAS

C. SAS

D. SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q4/5

Which corresponding side would make AAS apply if $\angle A \cong \angle D$ and $\angle C \cong \angle F$?

Side: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 15 • FORWARD 1 • Q5/5

SET 15 • HSG-CO.B.8

FORWARD 2

Prove congruence with ASA or AAS in overlapping triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Segments AD and BC cross at E. AB ∥ CD and E is the midpoint of AD. Prove that △ABE ≅ △DCE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q1/5

In $\triangle ABC$, the bisector of $\angle A$ is perpendicular to BC at D. Prove that AB = AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q2/5

In overlapping triangles ABE and ACD, $\angle A$ is shared, $AB = AC$, and $\angle B \cong \angle C$. Which criterion proves $\triangle ABE \cong \triangle ACD$?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q3/5

True or false?

An angle shared by two overlapping triangles cannot be used in a congruence proof, because it is the same angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q4/5

In △JKL and △JML, ∠KJL ≅ ∠MJL and ∠JLK ≅ ∠JLM. Which side completes ASA?

A. JL = JL

B. JK = JM

C. KL = ML

D. JK = KL

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S15-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 15 • FORWARD 2 • Q5/5

Explain SSS congruence with rigid motions

MAIN	Geometry main question	**HSG-CO.B.8**
BACKWARD 1	Three side lengths determine a triangle (triangle inequality)	**7.G.A.2**
BACKWARD 2	Find the points at given distances from two points (two circles)	**7.G.B.4**
BACKWARD 3	Rigid motions keep every length	**8.G.A.1**
FORWARD 1	Explain the Hypotenuse-Leg criterion for right triangles	**HSG-SRT.B.5**
FORWARD 2	Decide which criteria guarantee congruence, with counterexampl…	**HSG-SRT.B.5**

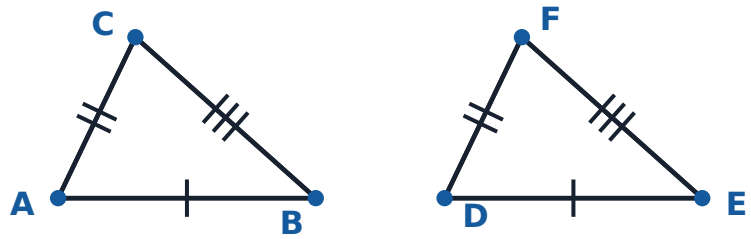
SET 16 • HSG-CO.B.8

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In $\triangle ABC$ and $\triangle DEF$, $AB = DE$, $BC = EF$, and $CA = FD$. Explain, using rigid motions, why $\triangle ABC \cong \triangle DEF$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q1/5

In the rigid-motion proof of SSS, A is on D, B is on E, and C′ (the image of C) is not F. Why is line DE the perpendicular bisector of C′F?

A. DE bisects ∠D

B. DE is parallel to C′F

C. D and E are each the same distance from C′ as from F

D. C′F is the longest segment

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q2/5

In the rigid-motion proof of SSS, line DE is the perpendicular bisector of C′F. Why does the reflection across DE carry C′ onto F, and why is that enough?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q3/5

The rigid-motion proof of SSS uses no angle information. Explain how the three side lengths alone locate the third vertex.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q4/5

True or false?

A triangle with sides 5, 7, and 9 is congruent to a triangle with sides 9, 5, and 7.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 16 • MAIN • Q5/5

SET 16 • HSG-CO.B.8

BACKWARD 1

Three side lengths determine a triangle (triangle inequality) • Grade 7 • Geometry • 7.G.A.2

True or false?

Side lengths 3, 4, and 8 can form a triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q1

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q1/5

True or false?

Side lengths 5, 6, and 10 form exactly one triangle (up to position).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q2

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q2/5

Which three lengths can form a triangle?

A. 5, 5, 11

B. 4, 5, 7

C. 2, 3, 6

D. 1, 1, 3

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q3

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q3/5

Two sides of a triangle are 6 and 10. Give the range of possible lengths for the third side.

Range: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q4

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q4/5

Can two different-shaped triangles both have sides 7, 8, and 9? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B1-Q5

Grade 7 • Geometry • 7.G.A.2 • Set 16 • BACKWARD 1 • Q5/5

SET 16 • HSG-CO.B.8

BACKWARD 2

Find the points at given distances from two points (two circles) • Grade 7 • Geometry • 7.G.B.4

Point P is 5 units from A and 5 units from B, and A and B are 6 units apart. How many such points P are there in the plane?

Number: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q1

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q1/5

True or false?

All points 3 cm from a point A form a circle with center A and radius 3 cm.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q2

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q2/5

Circles of radius 4 centered at A and B meet at two points. AB = 6. Where are the two points?

A. Both on line AB

B. One on each side of line AB

C. At A and B

D. Both on the same side of AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q3

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q3/5

True or false?

Two circles always meet in exactly two points.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q4

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q4/5

Points A and B are 10 apart. Is there a point 3 units from A and 4 units from B? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B2-Q5

Grade 7 • Geometry • 7.G.B.4 • Set 16 • BACKWARD 2 • Q5/5

SET 16 • HSG-CO.B.8

BACKWARD 3

Rigid motions keep every length • Grade 8 • Geometry • 8.G.A.1

Triangle ABC has sides 3, 5, and 7. After a rotation and a reflection, what are the side lengths of the image?

Lengths: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q1

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q1/5

True or false?

Under a rigid motion, the distance from the image of A to the image of C equals AC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q2

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q2/5

Which can a rigid motion change?

A. The perimeter

B. The length of a side

C. The area

D. The position of a triangle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q3

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q3/5

True or false?

A reflection doubles the distance from a point to the line of reflection.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q4

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q4/5

A point C is 6 units from D. After a reflection across a line through D, how far is the image of C from D?

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-B3-Q5

Grade 8 • Geometry • 8.G.A.1 • Set 16 • BACKWARD 3 • Q5/5

SET 16 • HSG-CO.B.8

FORWARD 1

Explain the Hypotenuse-Leg criterion for right triangles • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Right triangles ABC and DEF have right angles at C and F, hypotenuses AB = DE, and legs AC = DF. Prove that △ABC ≅ △DEF.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q1/5

Two right triangles each have hypotenuse 13 and one leg 5. Are they congruent? What is the other leg?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q2/5

True or false?

Hypotenuse-Leg is a special case of SSA that does guarantee congruence.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q3/5

Which information proves two right triangles congruent?

A. One pair of acute angles is equal

B. Their right angles are equal

C. Their hypotenuses are equal and one pair of legs is equal

D. Their hypotenuses are equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q4/5

In right triangles PQR and XYZ (right angles at R and Z), PQ = XY = 10 and QR = YZ = 6. Find PR and XZ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F1-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 1 • Q5/5

SET 16 • HSG-CO.B.8

FORWARD 2

Decide which criteria guarantee congruence, with counterexamples • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5

Explain with a counterexample why AAA is not a congruence criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q1/5

Explain with a counterexample why SSA is not a congruence criterion.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q2/5

Which list contains only valid triangle congruence criteria?

A. SSS, SAS, ASA, AAS

B. SSS, SSA, ASA, AAS

C. SSA, AAA, SAS, HL

D. SAS, AAA, ASA, SSS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q3

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q3/5

True or false?

If two triangles have the same three side lengths, they have the same three angle measures.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q4

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q4/5

True or false?

Two triangles with two pairs of congruent sides and equal areas must be congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S16-F2-Q5

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.B.5 • Set 16 • FORWARD 2 • Q5/5

Prove that vertical angles are congruent

MAIN	Geometry main question	**HSG-CO.C.9**
BACKWARD 1	Angles in a linear pair add to 180°	**7.G.B.5**
BACKWARD 2	Use properties of equality as reasons	**HSA-REI.A.1**
BACKWARD 3	Identify vertical angles and linear pairs in a figure	**7.G.B.5**
FORWARD 1	Use the vertical angle theorem in multistep proofs	**HSG-CO.C.9**
FORWARD 2	Prove statements about the bisectors of angles formed by two lin…	**HSG-CO.C.9**

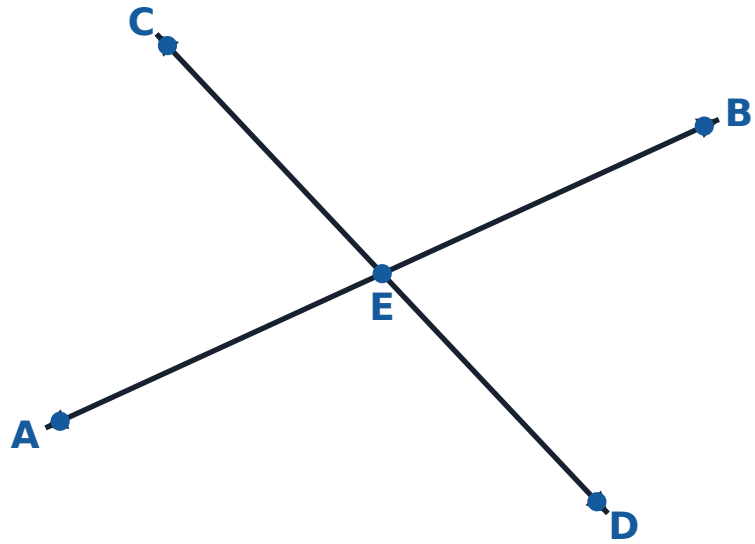
SET 17 • HSG-CO.C.9

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines AB and CD intersect at E. Prove that ∠AEC ≅ ∠BED.

Work space



Conclusion: _____

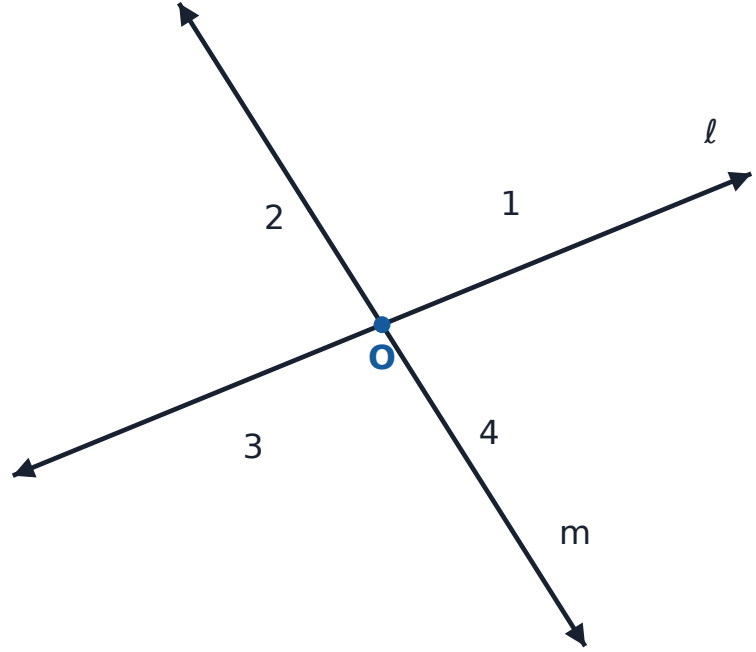
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q1/5

Lines ℓ and m meet at O, forming $\angle 1$, $\angle 2$, $\angle 3$, and $\angle 4$ in order around O. Prove that $\angle 2 \cong \angle 4$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q2/5

A proof that $\angle 1 \cong \angle 3$ for vertical angles 1 and 3 (with $\angle 2$ next to both):

1. $m\angle 1 + m\angle 2 = 180°$ (linear pair)

2. $m\angle 2 + m\angle 3 = 180°$ (linear pair)

3. $m\angle 1 + m\angle 2 = m\angle 2 + m\angle 3$ (?)

4. $m\angle 1 = m\angle 3$ (?)

Give the reasons for steps 3 and 4.

Reasons: _____

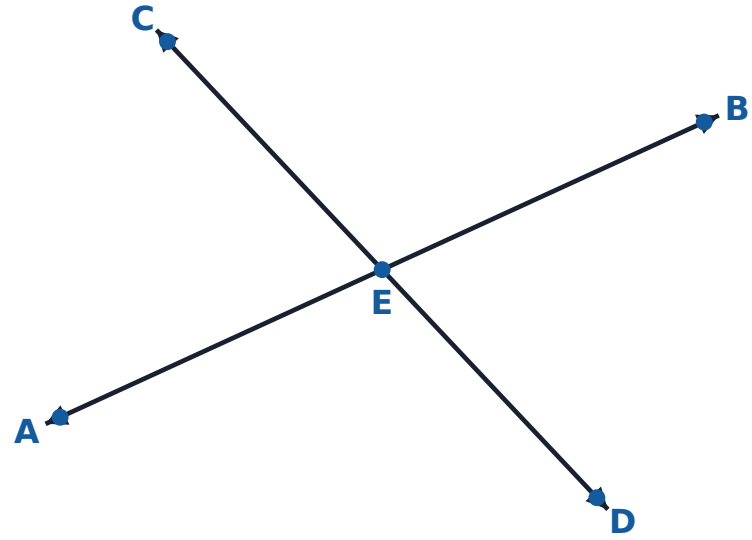
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q3/5

Lines AB and CD intersect at E. Explain, using a rigid motion, why $\angle AEC \cong \angle BED$.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q4/5

A student measures two vertical angles in one drawing, finds that both are 47°, and writes "vertical angles are congruent." Why is this not a proof?

A. Protractors are never accurate

B. Vertical angles are not always congruent

C. It checks one example; a proof must show the result for every pair of vertical angles

D. The student should have measured all four angles

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • MAIN • Q5/5

SET 17 • HSG-CO.C.9

BACKWARD 1

Angles in a linear pair add to 180° • Grade 7 • Geometry • 7.G.B.5

Two angles form a linear pair. One measures 128°. Find the other.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q1/5

∠PQR and ∠RQS form a linear pair, and m∠PQR = 3 · m∠RQS. Find m∠RQS.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q2/5

True or false?

The two angles of a linear pair are always supplementary.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q3/5

Points A, B, C are on a line with B between A and C, and ray BD is drawn. Which angles form a linear pair?

A. ∠DBC and ∠ABC

B. ∠BAD and ∠BCD

C. ∠ABD and ∠DBC

D. ∠ABD and ∠ABC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q4/5

True or false?

Two angles that add to 180° must form a linear pair.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B1-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 1 • Q5/5

SET 17 • HSG-CO.C.9

BACKWARD 2

Use properties of equality as reasons • High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1

From $m\angle 1 + 40° = m\angle 2 + 40°$, which property gives $m\angle 1 = m\angle 2$?

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q1

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q1/5

If $x + y = 180$ and $y + z = 180$, which property lets you write $x + y = y + z$?

Property: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q2

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q2/5

Which property justifies: if a = b, then b = a?

A. Reflexive property

B. Symmetric property

C. Distributive property

D. Transitive property

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q3

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q3/5

True or false?

The reflexive property says that any quantity equals itself, such as $m\angle A = m\angle A$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q4

High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q4/5

True or false?

From $2x = 2y$ you may conclude $x = y$ by the subtraction property of equality.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B2-Q5

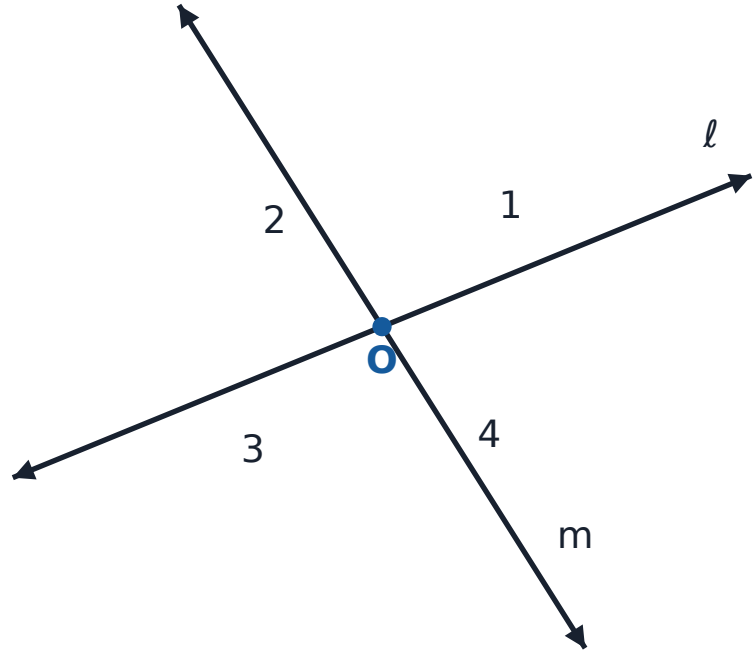
High School — Algebra I (course placement in this bank) • Algebra: Reasoning with Equations & Inequalities • HSA-REI.A.1 • Set 17 • BACKWARD 2 • Q5/5

SET 17 • HSG-CO.C.9

BACKWARD 3

Identify vertical angles and linear pairs in a figure • Grade 7 • Geometry • 7.G.B.5

Lines ℓ and m meet at O, forming $\angle 1$, $\angle 2$, $\angle 3$, $\angle 4$ in order. Which angle is vertical to $\angle 1$?



A. $\angle 3$

B. $\angle 1$

C. $\angle 4$

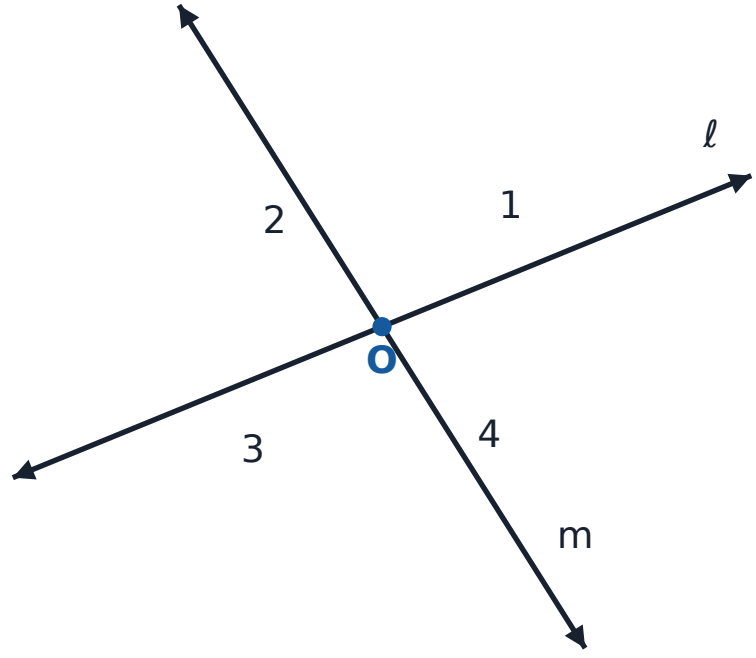
D. $\angle 2$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q1

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q1/5

Lines ℓ and m meet at O, forming $\angle 1$, $\angle 2$, $\angle 3$, $\angle 4$ in order. Name the two angles that form linear pairs with $\angle 2$.



Angles: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q2

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q2/5

True or false?

Vertical angles share a side.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q3

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q3/5

True or false?

When two lines cross, they form two pairs of vertical angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q4

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q4/5

Two lines cross. One angle is 63°. Find its vertical angle and the angles next to it.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-B3-Q5

Grade 7 • Geometry • 7.G.B.5 • Set 17 • BACKWARD 3 • Q5/5

SET 17 • HSG-CO.C.9

FORWARD 1

Use the vertical angle theorem in multistep proofs • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines AB and CD intersect at E, and ray EF bisects ∠AEC. Ray EG is the ray opposite ray EF. Prove that ray EG bisects ∠BED.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q1/5

Three lines meet at point O, forming six angles. Two of the angles that are next to each other are 50° and 70°. Prove that the six angles are 50°, 70°, 60°, 50°, 70°, 60° in order.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q2/5

Two lines meet. One angle is (4x + 10)° and its vertical angle is (6x − 30)°. Find x and all four angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q3/5

True or false?

If one angle formed by two crossing lines is a right angle, all four angles are right angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q4/5

Lines AB and CD meet at E with m∠AEC = 2x and m∠CEB = 3x. What is m∠BED?

A. 72°

B. 144°

C. 36°

D. 108°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 1 • Q5/5

SET 17 • HSG-CO.C.9

FORWARD 2

Prove statements about the bisectors of angles formed by two lines • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

∠ABD and ∠DBC form a linear pair. Ray BE bisects ∠ABD and ray BF bisects ∠DBC. Prove that ∠EBF is a right angle.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q1/5

Is the converse of the vertical angle theorem true: "If two angles are congruent, they are vertical angles"? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q2/5

Two lines meet at O. Prove that the bisectors of a pair of vertical angles lie on one line.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q3/5

True or false?

The bisectors of a pair of vertical angles are perpendicular.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q4/5

Two lines meet, and the bisectors of two adjacent angles are drawn. What angle do the bisectors form?

A. 45°

B. It depends on the lines

C. 180°

D. 90°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S17-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 17 • FORWARD 2 • Q5/5

Prove the angle theorems for parallel lines cut by a transversal

MAIN	Geometry main question	**HSG-CO.C.9**
BACKWARD 1	Name and use angles formed by parallel lines and a transversal	**8.G.A.5**
BACKWARD 2	Vertical angles and linear pairs as reasons	**HSG-CO.C.9**
BACKWARD 3	The parallel postulate and the definition of parallel lines	**HSG-CO.A.1**
FORWARD 1	Prove the converse: congruent alternate interior angles make line...	**HSG-CO.C.9**
FORWARD 2	Use parallel-line theorems to prove the triangle angle sum	**HSG-CO.C.10**

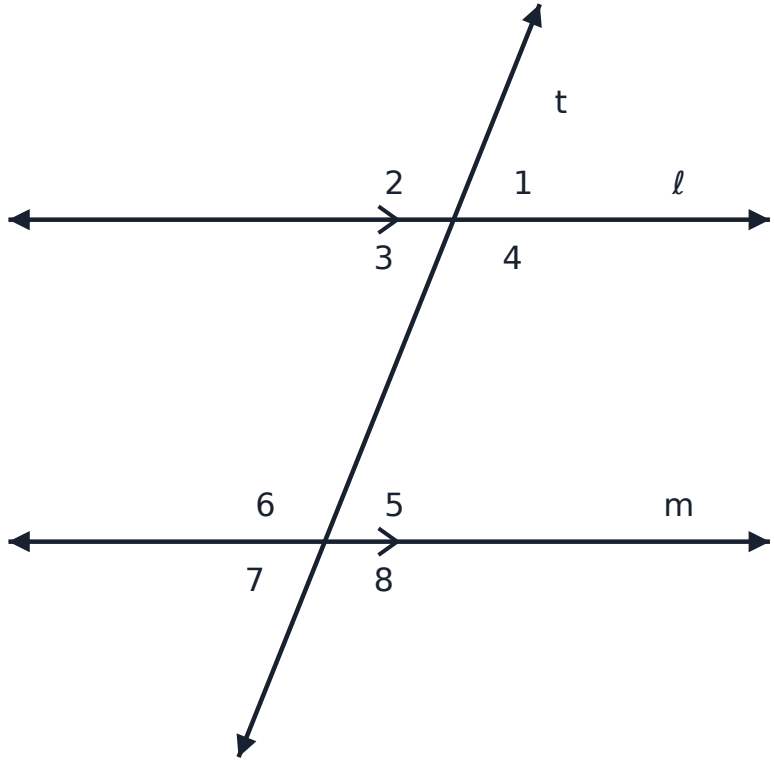
SET 18 • HSG-CO.C.9

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are parallel and are cut by transversal t. Assume the corresponding angles postulate (corresponding angles formed by a transversal of parallel lines are congruent). Prove that alternate interior angles $\angle 3$ and $\angle 5$ are congruent.

Work space



Conclusion: _____

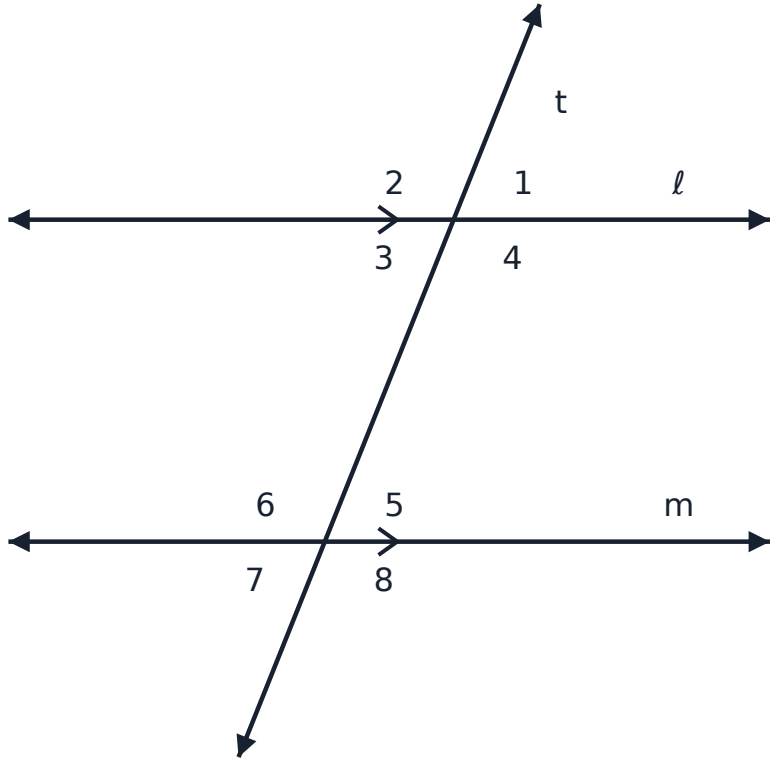
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q1/5

Lines ℓ and m are parallel and are cut by transversal t. Assume that alternate interior angles are congruent. Prove that corresponding angles $\angle 2$ and $\angle 6$ are congruent.

Work space



Conclusion: _____

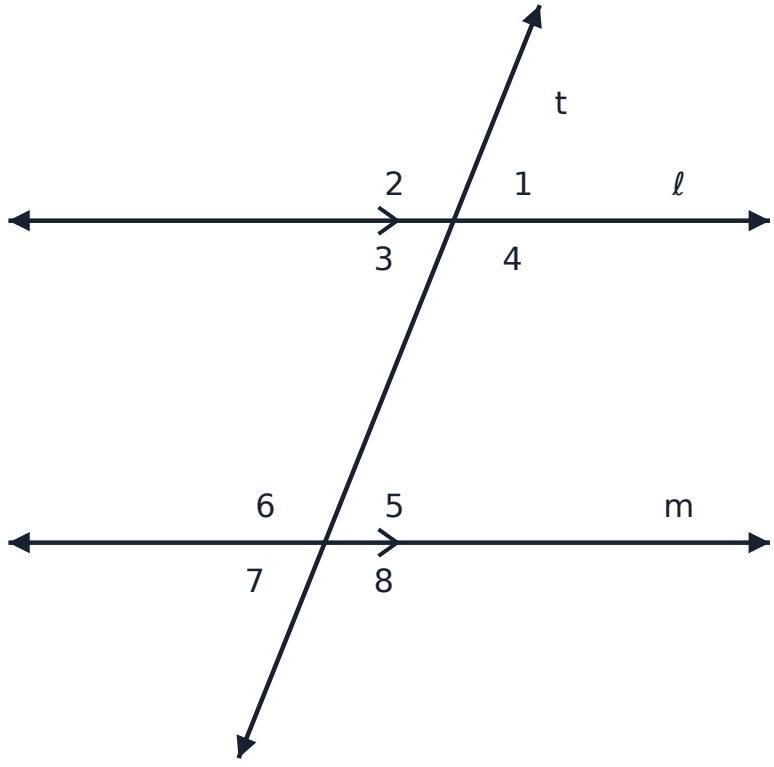
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q2/5

Lines ℓ and m are parallel and are cut by transversal t, crossing ℓ at P and m at Q. Explain, using a translation, why corresponding angles $\angle 1$ and $\angle 5$ are congruent.

Work space



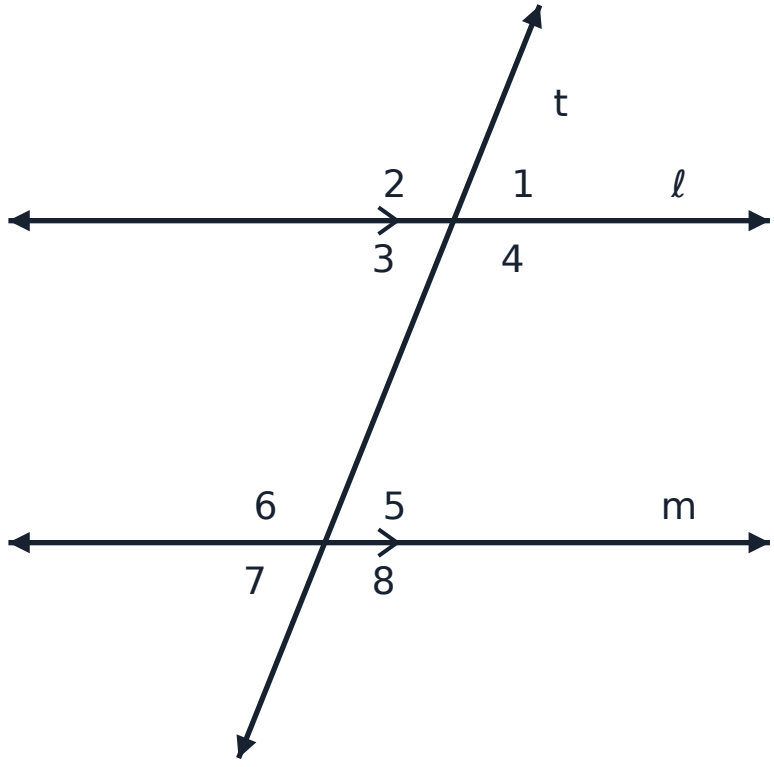
Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q3/5

Lines $\ell \parallel m$ are cut by transversal t, and $m\angle 1 = 115°$. Find $m\angle 5$, $m\angle 3$, and $m\angle 6$. Give a reason for each.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q4/5

A student "proves" that alternate interior angles are congruent by saying: "The lines are parallel because the alternate interior angles are congruent." What is wrong?

A. Nothing; the argument is correct

B. The argument is circular: it assumes the conclusion

C. Alternate interior angles are supplementary

D. The lines must be perpendicular

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-M-Q5

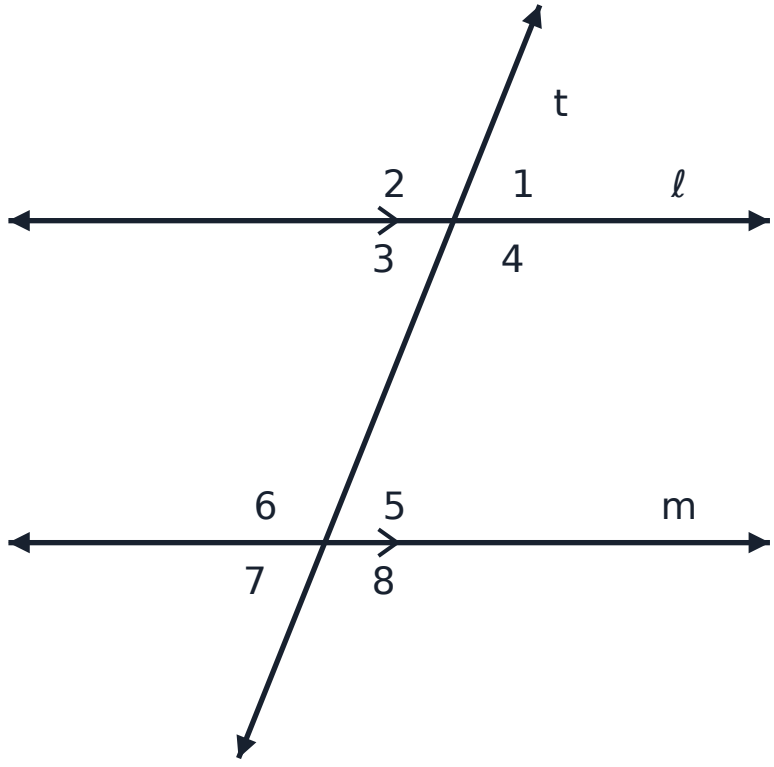
High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • MAIN • Q5/5

SET 18 • HSG-CO.C.9

BACKWARD 1

Name and use angles formed by parallel lines and a transversal • Grade 8 • Geometry • 8.G.A.5

Lines $\ell \parallel m$ are cut by transversal t. $\angle 4$ measures $70°$. Find the measure of $\angle 6$.



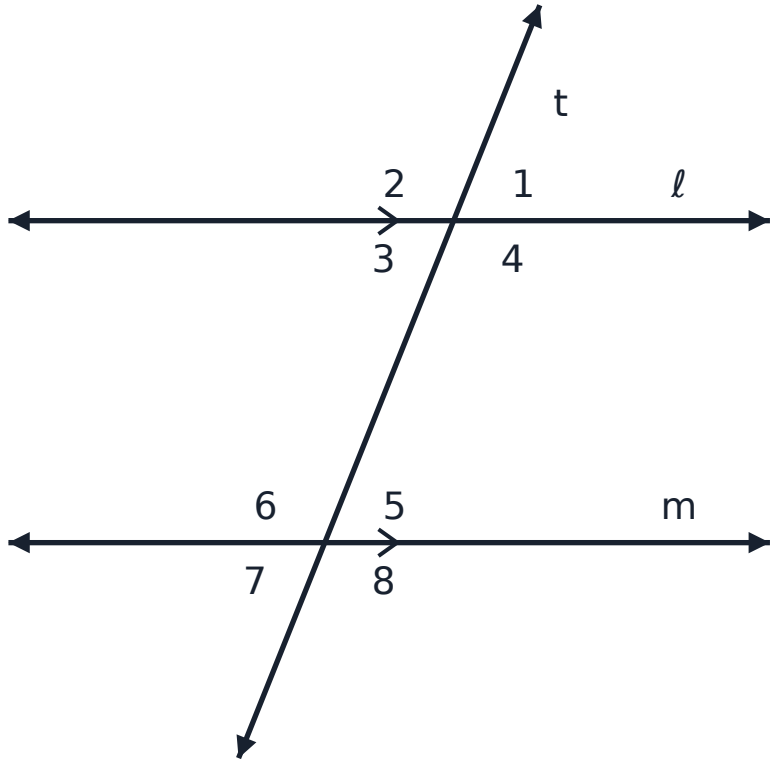
Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q1/5

Lines $\ell \parallel m$ are cut by transversal t. Which angle corresponds to $\angle 2$?



A. $\angle 7$

B. $\angle 8$

C. $\angle 6$

D. $\angle 5$

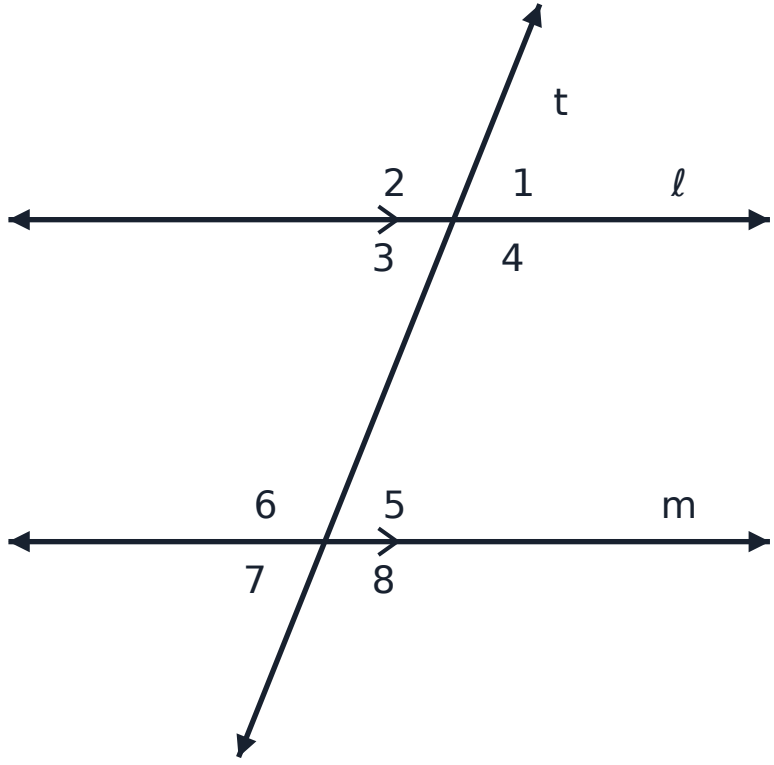
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q2/5

True or false?

Lines $\ell \parallel m$ are cut by transversal t. $\angle 4$ and $\angle 5$ are same-side interior angles, so they add to 180°.



A. True **B.** False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q3/5

True or false?

When two lines are cut by a transversal, corresponding angles are congruent even if the lines are not parallel.

A. True

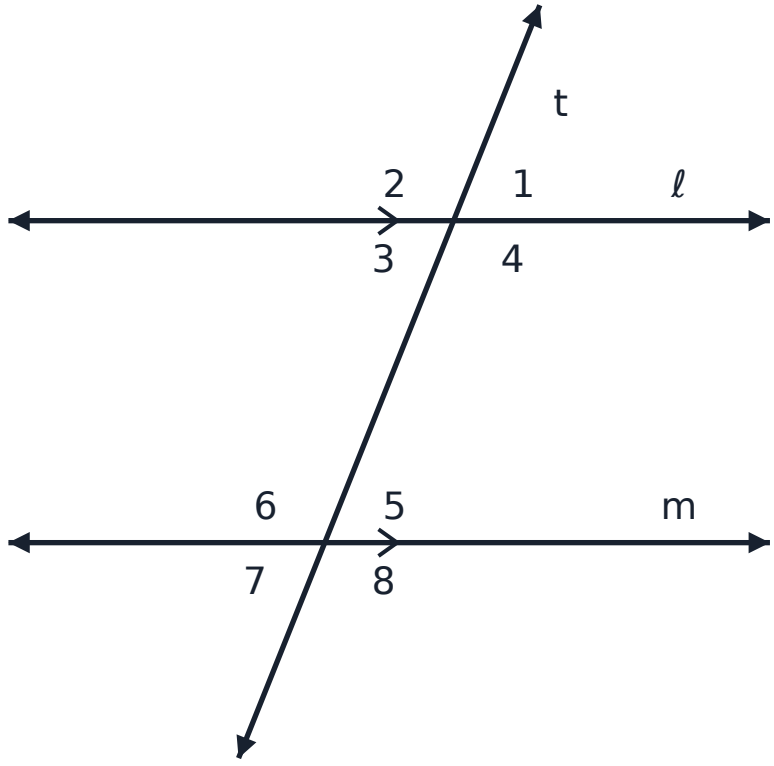
B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q4/5

Lines $\ell \parallel m$ are cut by transversal t. $\angle 8$ measures 112°. Find the measure of $\angle 2$.



Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 18 • BACKWARD 1 • Q5/5

SET 18 • HSG-CO.C.9

BACKWARD 2

Vertical angles and linear pairs as reasons • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Two lines cross. Name the reason two opposite angles are congruent.

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q1/5

Name the reason two adjacent angles on a line add to 180°.

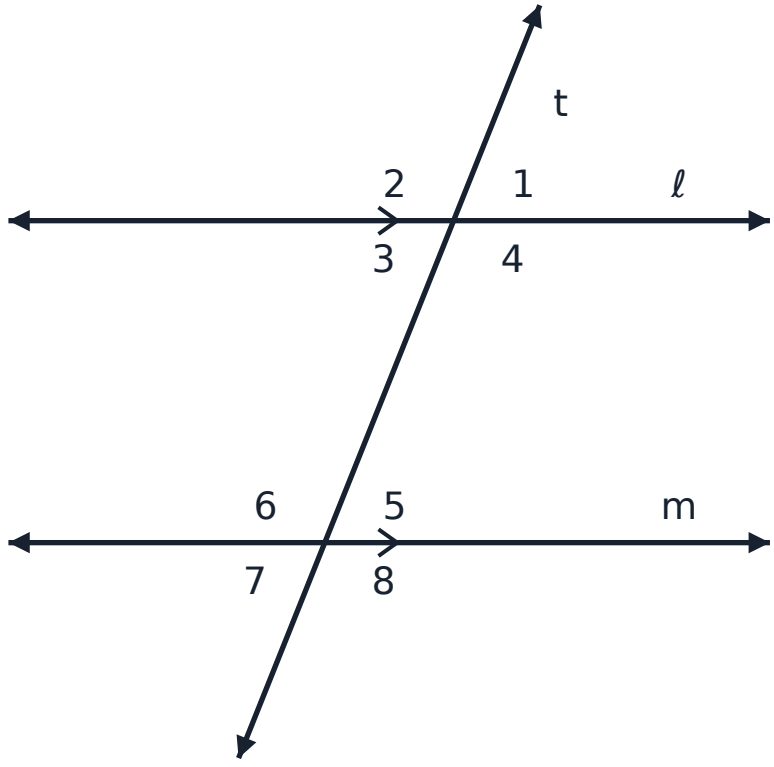
Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q2/5

At the top crossing, $\angle 1 = 110°$. What is $\angle 2$ and why?



A. 110°, linear pair

B. 70°, linear pair

C. 110°, vertical angles

D. 70°, vertical angles

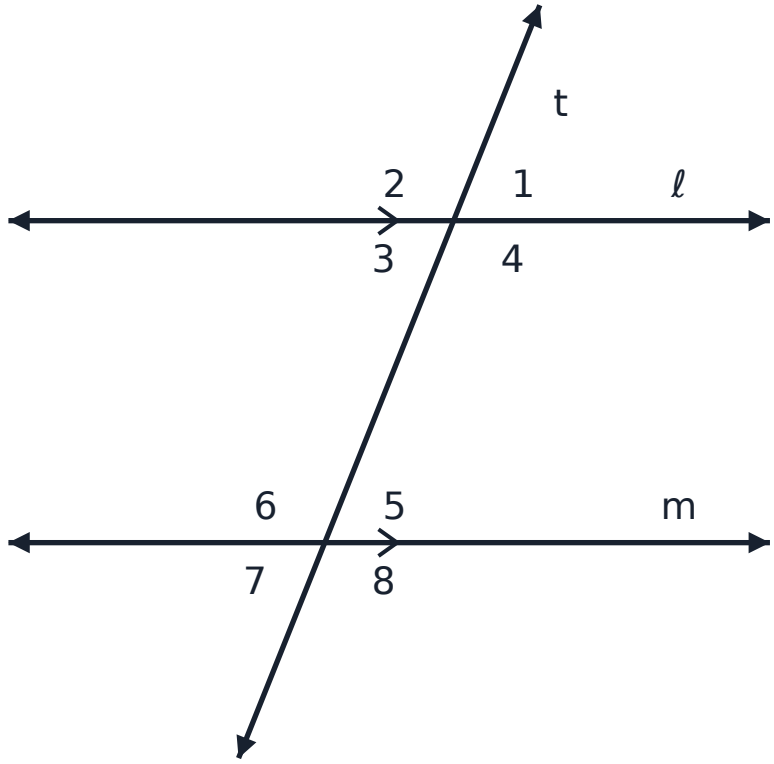
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q3/5

True or false?

Angles 1 and 3 are vertical angles in the figure.



A. True

B. False

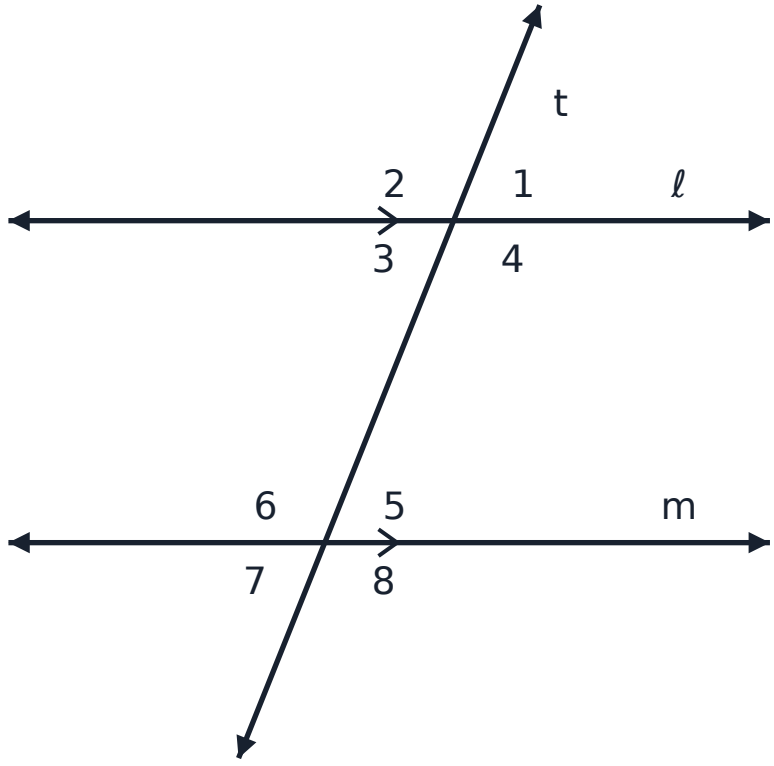
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q4/5

True or false?

Angles 1 and 4 are vertical angles in the figure.



A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • BACKWARD 2 • Q5/5

SET 18 • HSG-CO.C.9

BACKWARD 3

The parallel postulate and the definition of parallel lines • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

True or false?

Through a point not on a line, there is exactly one line parallel to it (in a plane).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q1/5

Line ℓ is parallel to line m. Line n passes through a point of ℓ and is also parallel to m. What can you conclude about n?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q2/5

Which pair of lines in a plane are parallel?

A. $x = 2$ and $y = 2$

B. $y = x$ and $y = -x$

C. $y = 2x + 1$ and $y = -2x + 1$

D. $y = 2x + 1$ and $y = 2x - 5$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q3/5

True or false?

Two lines in a plane that never meet are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q4/5

Line $\ell \parallel$ line m. A translation maps m to a line through a point of ℓ. Must that image line be parallel to m or the same as m?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION ID S18-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 18 • BACKWARD 3 • Q5/5

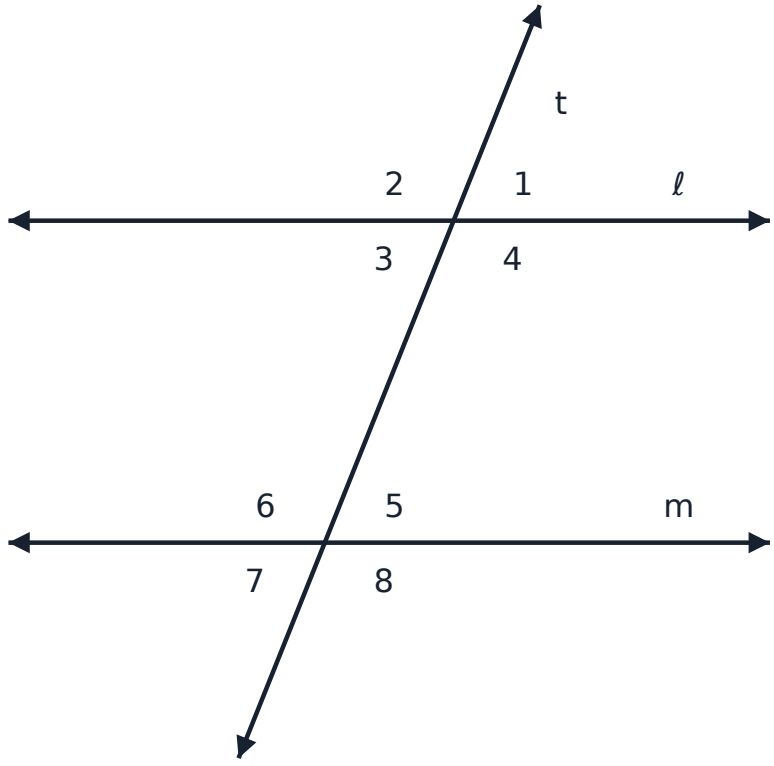
SET 18 • HSG-CO.C.9

FORWARD 1

Prove the converse: congruent alternate interior angles make lines parallel • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are cut by transversal t, and alternate interior angles $\angle 3$ and $\angle 5$ are congruent. Prove that $\ell \parallel$ m.

Work space



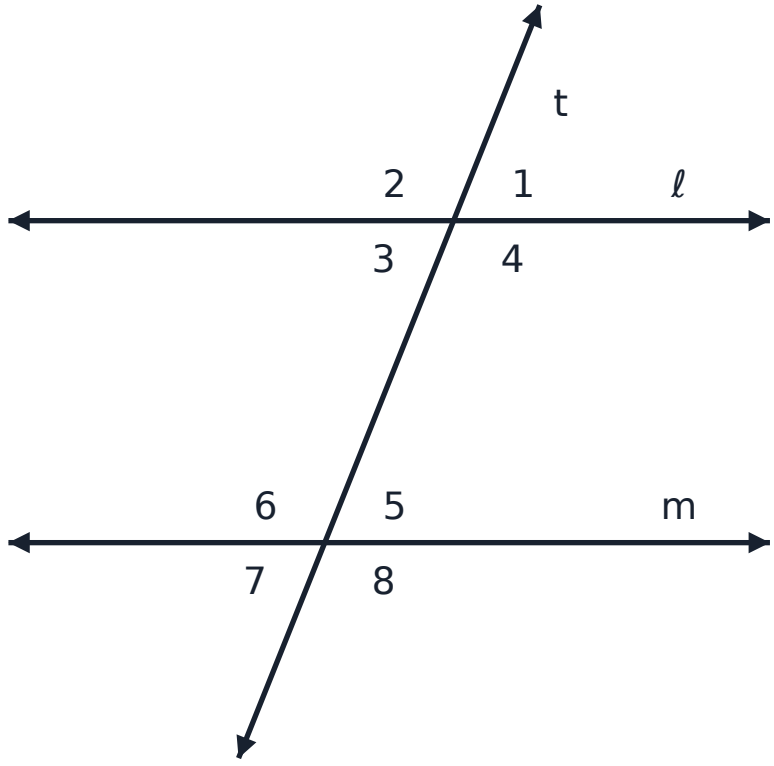
Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q1/5

Lines ℓ and m are cut by transversal t with $m\angle 4 = (5x - 20)°$ and $m\angle 6 = (3x + 30)°$. For what value of x is $\ell \parallel m$? Explain.



Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q2/5

True or false?

If same-side interior angles are supplementary, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q3/5

Which fact proves that lines ℓ and m are parallel?

A. A pair of corresponding angles is congruent

B. A pair of vertical angles is congruent

C. A linear pair adds to 180°

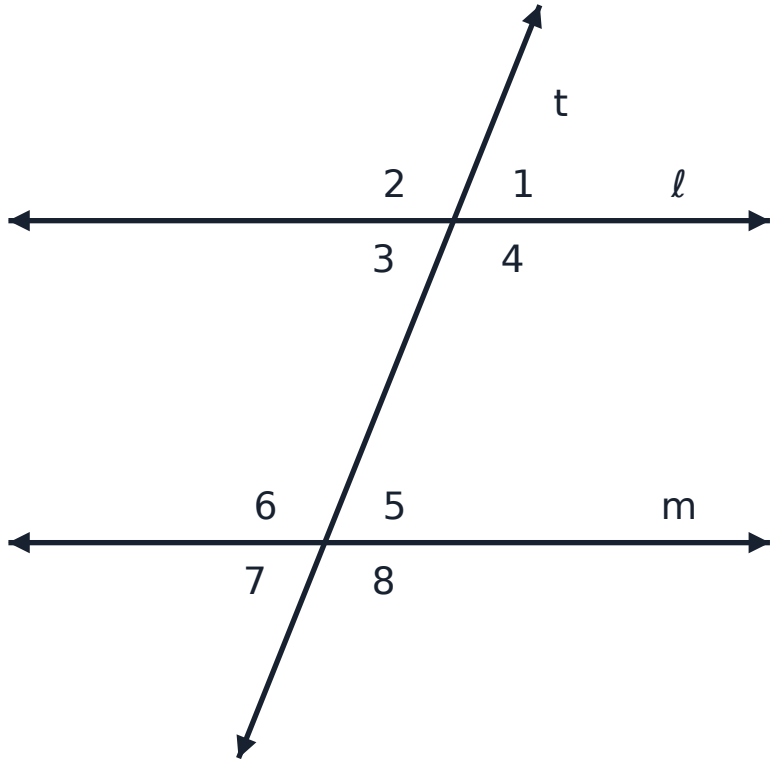
D. Two angles at the same crossing are equal

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 18 • FORWARD 1 • Q4/5

Lines ℓ and m are cut by a transversal. $m\angle 1 = 72°$ and $m\angle 7 = 108°$. Are ℓ and m parallel? Explain.



Answer: _____

SET 18 • HSG-CO.C.9

FORWARD 2

Use parallel-line theorems to prove the triangle angle sum • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that the angles of triangle ABC add to 180°. Begin by drawing the line through B parallel to AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q1/5

In triangle ABC, a line through B parallel to AC makes a 50° angle with BA (on A's side) and a 65° angle with BC (on C's side). Find all three angles of the triangle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q2/5

True or false?

The proof of the triangle angle sum works without any fact about parallel lines.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q3/5

Which theorem is used to show the new angles at B equal ∠A and ∠C?

A. Alternate interior angles theorem

B. Isosceles triangle theorem

C. Triangle inequality

D. Vertical angles theorem

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q4/5

Explain why the exterior angle at C of triangle ABC equals m∠A + m∠B.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S18-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 18 • FORWARD 2 • Q5/5

SET 19 • HSG-CO.C.9 • CONGRUENCE: PROVING THEOREMS

Prove the perpendicular bisector theorem and its converse

MAIN	Geometry main question	**HSG-CO.C.9**
BACKWARD 1	Use the definitions of midpoint and perpendicular bisector	**HSG-CO.A.1**
BACKWARD 2	Use SAS and SSS to prove triangles congruent	**HSG-CO.B.8**
BACKWARD 3	Corresponding parts of congruent triangles and right angles from…	**HSG-CO.B.7**
FORWARD 1	Use the perpendicular bisector to find points equidistant from two…	**HSG-GPE.B.5**
FORWARD 2	Prove that the perpendicular bisectors of a triangle meet at one p…	**HSG-C.A.3**

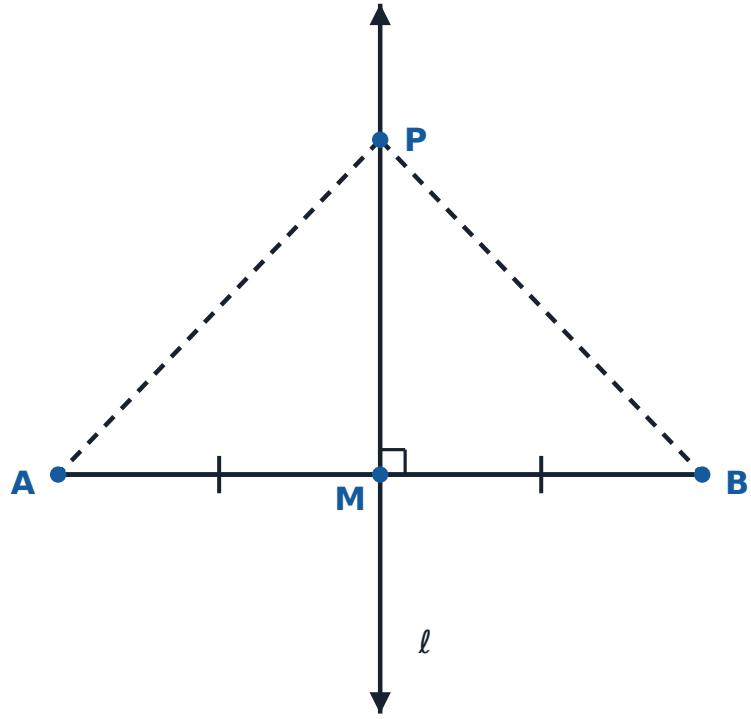
SET 19 • HSG-CO.C.9

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Line ℓ is the perpendicular bisector of segment AB, meeting AB at M. Point P is on ℓ. Prove that PA = PB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q1/5

Point P is not on line AB, and PA = PB. Let M be the midpoint of AB. Prove that line PM is perpendicular to AB, so P is on the perpendicular bisector of AB.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q2/5

The theorem says the points on the perpendicular bisector of AB are EXACTLY the points equidistant from A and B. Which two statements must be proved?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q3/5

In a coordinate plane, A(−a, 0) and B(a, 0) with a > 0. Prove that every point P(0, y) on the y-axis is equidistant from A and B, and that every point P(x, y) with PA = PB has x = 0.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q4/5

A student proves only that every point on the perpendicular bisector of AB is equidistant from A and B. What is missing?

A. A proof that the bisector is a line

B. Nothing; one direction proves "exactly"

C. The converse: every point equidistant from A and B is on the perpendicular bisector

D. A measurement of PA and PB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 19 • MAIN • Q5/5

SET 19 • HSG-CO.C.9

BACKWARD 1

Use the definitions of midpoint and perpendicular bisector • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

M is the midpoint of AB and AB = 14. Find AM.

AM = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q1/5

True or false?

The perpendicular bisector of AB passes through the midpoint of AB and is perpendicular to AB.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q2/5

Line ℓ meets AB at M with AM = MB, but ℓ is not perpendicular to AB. What is ℓ?

A. Not related to AB

B. Parallel to AB

C. The perpendicular bisector of AB

D. A bisector of AB that is not the perpendicular bisector

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q3/5

Line ℓ is perpendicular to AB at M, and AM = 3, MB = 5. Is ℓ the perpendicular bisector of AB? Explain.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q4/5

True or false?

A segment has exactly one perpendicular bisector in a plane.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 19 • BACKWARD 1 • Q5/5

SET 19 • HSG-CO.C.9

BACKWARD 2

Use SAS and SSS to prove triangles congruent • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

In $\triangle PMA$ and $\triangle PMB$, PM is shared, AM = BM, and $\angle PMA \cong \angle PMB$. Which criterion proves they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q1/5

In $\triangle PMA$ and $\triangle PMB$, $PA = PB$, $AM = BM$, and PM is shared. Which criterion proves they are congruent?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q2/5

Two triangles share a side and have two other pairs of congruent sides. Which criterion applies?

A. SAS

B. ASA

C. SSS

D. AAS

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q3/5

True or false?

A side shared by two triangles counts as a pair of congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q4/5

True or false?

SSA is a valid congruence criterion.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 19 • BACKWARD 2 • Q5/5

SET 19 • HSG-CO.C.9

BACKWARD 3

Corresponding parts of congruent triangles and right angles from a linear pair • High School — Geometry • Geometry: Congruence • HSG-CO.B.7

△PMA ≅ △PMB. Which angle corresponds to ∠PMA?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q1/5

Two congruent angles form a linear pair. What is each angle's measure?

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q2/5

True or false?

If $\triangle PMA \cong \triangle PMB$, then $PA = PB$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q3/5

$\triangle XYZ \cong \triangle XWZ$. Which pair of segments must be congruent?

A. YZ and XW

B. YZ and WZ

C. XY and XZ

D. XZ and YW

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q4/5

True or false?

Two angles in a linear pair are always congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.7 • Set 19 • BACKWARD 3 • Q5/5

SET 19 • HSG-CO.C.9

FORWARD 1

Use the perpendicular bisector to find points equidistant from two points • High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5

Find an equation of the perpendicular bisector of the segment from A(2, 1) to B(6, 5).

Equation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q1

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q1/5

Find the point on the x-axis that is equidistant from A(2, 1) and B(6, 5).

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q2

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q2/5

True or false?

The point (4, 4) is equidistant from (0, 0) and (8, 6).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q3

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q3/5

Which point is equidistant from A(0, 0) and B(4, 0)?

A. (3, 1)

B. (4, 4)

C. (2, 7)

D. (0, 2)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q4

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q4/5

Points A(−1, 2) and B(3, 2). Describe all points equidistant from A and B.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F1-Q5

High School — Geometry • Geometry: Expressing Geometric Properties with Equations • HSG-GPE.B.5 • Set 19 • FORWARD 1 • Q5/5

SET 19 • HSG-CO.C.9

FORWARD 2

Prove that the perpendicular bisectors of a triangle meet at one point • High School — Geometry • Geometry: Circles • HSG-C.A.3

Prove that the perpendicular bisectors of the sides of triangle ABC meet at one point O, and that O is equidistant from A, B, and C.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q1

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q1/5

Triangle ABC has A(0, 0), B(6, 0), C(0, 8). Find the point equidistant from all three vertices.

Point: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q2

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q2/5

True or false?

The point where the perpendicular bisectors of a triangle meet is always inside the triangle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q3

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q3/5

Three towns are to share a well equally distant from all three. Where should it go?

A. Where the perpendicular bisectors of the triangle's sides meet

B. Where the angle bisectors meet

C. At the midpoint of the longest side

D. Where the medians meet

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q4

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q4/5

Explain why showing that the perpendicular bisectors of AB and BC meet at O is enough to show that all three meet at O.

Explanation: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S19-F2-Q5

High School — Geometry • Geometry: Circles • HSG-C.A.3 • Set 19 • FORWARD 2 • Q5/5

Prove that the angles of a triangle add to 180°

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Use the angle sum and parallel-line angles informally	**8.G.A.5**
BACKWARD 2	Angles that form a straight angle add to 180°	**4.MD.C.7**
BACKWARD 3	Alternate interior angles of parallel lines are congruent	**HSG-CO.C.9**
FORWARD 1	Prove the exterior angle theorem	**HSG-CO.C.10**
FORWARD 2	Generalize the angle sum to polygons	**HSG-CO.C.10**

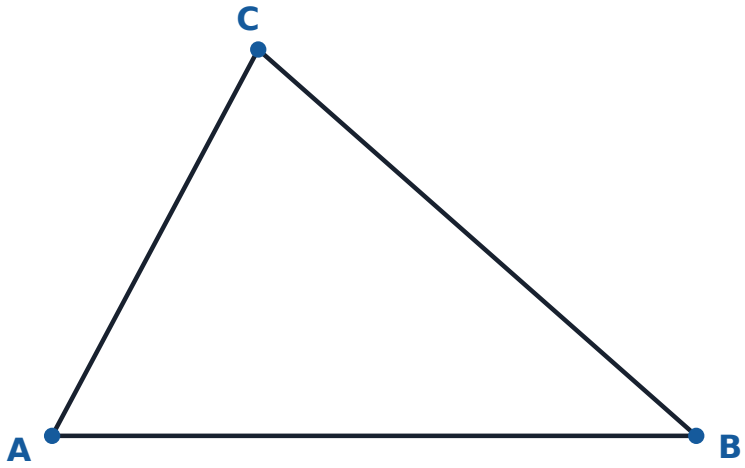
SET 20 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that
$m\angle A + m\angle B + m\angle C = 180°$ in triangle ABC. Draw the line through C parallel to AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q1/5

In triangle PQR, draw the line through P parallel to QR. Prove that the angles of triangle PQR add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q2/5

In triangle ABC, extend side BC past C to point D, and draw ray CE parallel to BA (E on the same side of line BD as A). Prove that the angles of triangle ABC add to 180°.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q3/5

A proof of the triangle angle sum (line ℓ through C parallel to AB, making $\angle 1$ next to CA and $\angle 2$ next to CB):

1. $m\angle 1 + m\angle ACB + m\angle 2 = 180°$ (?)

2. $\angle 1 \cong \angle A$ and $\angle 2 \cong \angle B$ (?)

3. $m\angle A + m\angle ACB + m\angle B = 180°$ (?)

Give the three reasons.

Reasons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q4/5

A student proves the angle sum theorem using "an exterior angle equals the sum of the two remote interior angles." What is wrong?

A. Exterior angles do not exist in triangles

B. Nothing is wrong

C. That theorem is usually proved FROM the angle sum, so the argument is circular

D. The exterior angle theorem is false

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • MAIN • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 1

Use the angle sum and parallel-line angles informally • Grade 8 • Geometry • 8.G.A.5

Two angles of a triangle are 47° and 68°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q1

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q1/5

True or false?

Tearing off the three corners of a paper triangle and lining them up makes a straight angle.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q2

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q2/5

A triangle has angles x, 2x, and 3x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q3

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q3/5

Lines $\ell \parallel m$ are cut by a transversal. An angle at ℓ is 72°. Which angle at m must also be 72°?

A. Its alternate interior angle

B. Its same-side interior angle

C. Every angle at m

D. The angle next to it on the same line

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q4

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q4/5

True or false?

A triangle can have two obtuse angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B1-Q5

Grade 8 • Geometry • 8.G.A.5 • Set 20 • BACKWARD 1 • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 2

Angles that form a straight angle add to 180° • Grade 4 • Measurement & Data • 4.MD.C.7

Three angles together form a straight angle. Two are 40° and 75°. Find the third.

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q1

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q1/5

True or false?

The angles around a point on one side of a straight line add to 180°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q2

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q2/5

$\angle 1$, $\angle 2$, and $\angle 3$ form a straight angle, with $\angle 1 = \angle 3 = 50°$. Find $\angle 2$.

A. 50°

B. 100°

C. 80°

D. 130°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q3

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q3/5

Angle ABC is a straight angle. Ray BD makes ∠ABD = 3x and ∠DBC = 2x. Find x.

x = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q4

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q4/5

True or false?

Two angles that make up a straight angle must be equal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B2-Q5

Grade 4 • Measurement & Data • 4.MD.C.7 • Set 20 • BACKWARD 2 • Q5/5

SET 20 • HSG-CO.C.10

BACKWARD 3

Alternate interior angles of parallel lines are congruent • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines $\ell \parallel m$ are cut by transversal t. An alternate interior angle of a 58° angle measures

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q1/5

True or false?

Alternate interior angles formed by a transversal of parallel lines are congruent.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q2/5

Line ℓ through C is parallel to AB. Which angle at C is congruent to $\angle A$?

A. The angle between CA and CB

B. The angle between ℓ and CB

C. The angle between ℓ and CA

D. $\angle ACB$

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q3/5

Why is it allowed to draw the line through C parallel to AB?

Reason: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q4/5

True or false?

Alternate interior angles are congruent for any two lines cut by a transversal.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-B3-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 20 • BACKWARD 3 • Q5/5

SET 20 • HSG-CO.C.10

FORWARD 1

Prove the exterior angle theorem • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, side BC is extended past C to D. Prove that $m\angle ACD = m\angle A + m\angle B$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q1/5

An exterior angle of a triangle measures 125°. One remote interior angle is 48°. Find the other remote interior angle and the adjacent interior angle.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q2/5

True or false?

An exterior angle of a triangle is always greater than 90°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q3/5

The exterior angle at C is (5x + 10)° and the remote interior angles are (2x + 15)° and (2x + 20)°. Find x.

A. 5

B. 35

C. 25

D. 15

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q4/5

Prove that the three exterior angles of a triangle (one at each vertex) add to 360°.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 1 • Q5/5

SET 20 • HSG-CO.C.10

FORWARD 2

Generalize the angle sum to polygons • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Prove that the interior angles of any convex polygon with n sides add to $(n - 2) \cdot 180°$.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q1/5

Find the measure of each interior angle of a regular octagon.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q2/5

Each interior angle of a regular polygon is 156°. How many sides does it have?

Sides: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q3/5

True or false?

The interior angles of a hexagon add to 1,080°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q4/5

A pentagon has angles 100°, 110°, 120°, 95°, and x°. Find x.

A. 125

B. 115

C. 135

D. 105

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S20-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 20 • FORWARD 2 • Q5/5

SET 21 • HSG-CO.C.10 • CONGRUENCE: PROVING THEOREMS

Prove that the base angles of an isosceles triangle are congruent

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Prove triangles congruent with SAS or SSS	**HSG-CO.B.8**
BACKWARD 2	Use the definitions of angle bisector and midpoint	**HSG-CO.A.1**
BACKWARD 3	Classify triangles and name legs, base, and base angles	**4.G.A.2**
FORWARD 1	Prove the converse: congruent base angles give congruent sides	**HSG-CO.C.10**
FORWARD 2	Use isosceles triangle theorems in multistep problems and proofs	**HSG-CO.C.10**

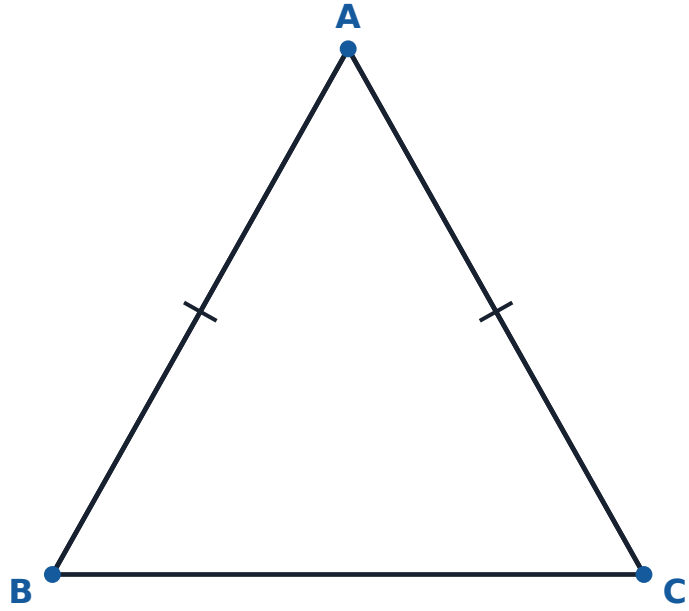
SET 21 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C. (Draw the bisector of ∠A, meeting BC at D.)

Work space



Conclusion: _____

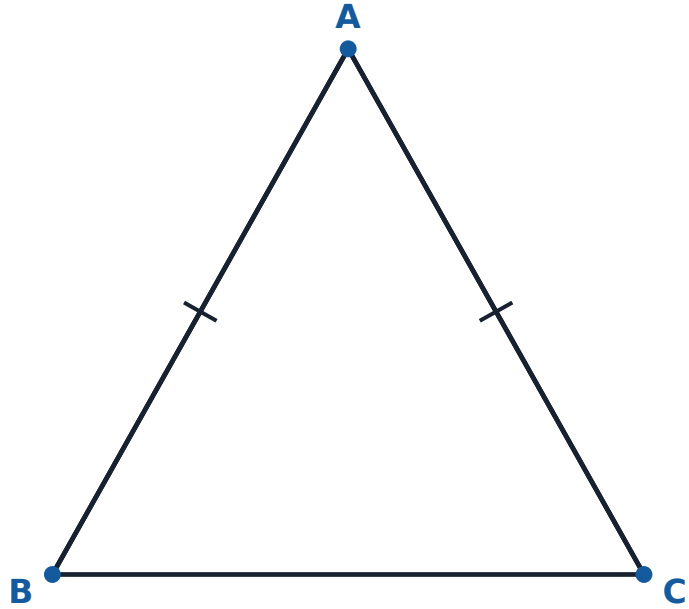
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q1/5

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C using the midpoint M of BC.

Work space



Conclusion: _____

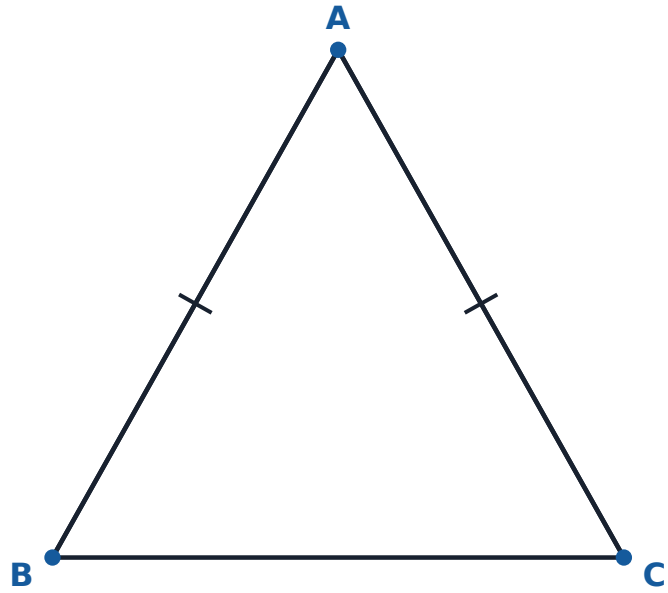
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q2/5

In triangle ABC, AB = AC. Prove that ∠B ≅ ∠C using a rigid motion: the reflection across the bisector of ∠A.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q3/5

Prove that every equilateral triangle is equiangular (all three angles are congruent).

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q4/5

In the proof that base angles of isosceles triangle ABC (AB = AC) are congruent, the bisector of ∠A meets BC at D. Which criterion proves △ABD ≅ △ACD?

A. SAS

B. SSS

C. AAS

D. ASA

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • MAIN • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 1

Prove triangles congruent with SAS or SSS • High School — Geometry • Geometry: Congruence • HSG-CO.B.8

Two triangles share side AD, and AB = AC, ∠BAD ≅ ∠CAD. Which criterion shows △ABD ≅ △ACD?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q1/5

Two triangles share side AM, and AB = AC, BM = CM. Which criterion shows $\triangle ABM \cong \triangle ACM$?

Criterion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q2/5

True or false?

A shared side counts as a pair of congruent sides in a congruence proof.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q3/5

Which information is NOT enough for congruence?

A. Three sides

B. Two sides and the included angle

C. Two angles and the included side

D. Two sides and a non-included angle

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q4/5

True or false?

If $\triangle ABD \cong \triangle ACD$, then $\angle ABD \cong \angle ACD$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.B.8 • Set 21 • BACKWARD 1 • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 2

Use the definitions of angle bisector and midpoint • High School — Geometry • Geometry: Congruence • HSG-CO.A.1

Ray AD bisects ∠BAC, and m∠BAC = 64°. Find m∠BAD.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q1/5

M is the midpoint of BC, and BC = 18. Find BM.

BM = _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q2/5

True or false?

An angle bisector divides an angle into two congruent angles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q3/5

Ray AD bisects ∠BAC. Which equation is true?

A. m∠BAD = m∠BAC

B. m∠BAD = m∠DAC

C. AD = AB

D. BD = DC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q4/5

True or false?

The bisector of $\angle A$ of triangle ABC always passes through the midpoint of BC.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.A.1 • Set 21 • BACKWARD 2 • Q5/5

SET 21 • HSG-CO.C.10

BACKWARD 3

Classify triangles and name legs, base, and base angles • Grade 4 • Geometry • 4.G.A.2

Triangle PQR has PQ = PR. Name the base and the base angles.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q1

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q1/5

True or false?

An isosceles triangle has at least two congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q2

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q2/5

Which triangle is scalene?

A. Sides 3, 3, 3

B. Sides 5, 5, 5

C. Sides 4, 5, 6

D. Sides 4, 4, 6

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q3

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q3/5

In isosceles triangle XYZ with XY = XZ, which angle is the vertex angle?

Angle: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q4

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q4/5

True or false?

Every equilateral triangle is isosceles.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-B3-Q5

Grade 4 • Geometry • 4.G.A.2 • Set 21 • BACKWARD 3 • Q5/5

SET 21 • HSG-CO.C.10

FORWARD 1

Prove the converse: congruent base angles give congruent sides • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, $\angle B \cong \angle C$. Prove that AB = AC.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q1/5

In triangle PQR, m∠Q = 70° and m∠R = 70°, and PQ = 3x + 2, PR = 5x − 6. Find x and PQ.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q2/5

True or false?

A triangle with two congruent angles must have two congruent sides.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q3/5

Triangle ABC has $m\angle A = 50°$ and $m\angle B = 65°$. Which two sides are congruent?

A. No two sides

B. AB and BC

C. AB and AC

D. AC and BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q4/5

Prove that every equiangular triangle is equilateral.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 1 • Q5/5

SET 21 • HSG-CO.C.10

FORWARD 2

Use isosceles triangle theorems in multistep problems and proofs • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

In triangle ABC, AB = AC and the vertex angle ∠A measures 38°. Find m∠B.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q1/5

Point D is on side BC of triangle ABC with AD = BD = CD. Find m∠BAC.

Measure: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q2/5

In triangle ABC, AB = AC, and D and E are points on BC with BD = CE. Prove that AD = AE.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q3/5

True or false?

If the vertex angle of an isosceles triangle is 100°, each base angle is 80°.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q4/5

In isosceles triangle PQR with PQ = PR, the exterior angle at Q (on the extension of RQ past Q) is 112°. Find m∠P.

A. 56°

B. 112°

C. 68°

D. 44°

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S21-F2-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 21 • FORWARD 2 • Q5/5

SET 22 • HSG-CO.C.10 • CONGRUENCE: PROVING THEOREMS

Prove the midsegment theorem

MAIN	Geometry main question	**HSG-CO.C.10**
BACKWARD 1	Find midpoints of segments in the coordinate plane	**6.G.A.3**
BACKWARD 2	Find and compare slopes to decide whether lines are parallel	**8.EE.B.6**
BACKWARD 3	Find lengths with the distance formula	**8.G.B.8**
BACKWARD 4	Prove lines parallel from congruent alternate interior angles	**HSG-CO.C.9**
FORWARD 1	Prove that the midpoints of a quadrilateral's sides form a parallelo	**HSG-CO.C.11**
FORWARD 2	Midsegment triangles: similarity, scale factor, and area	**HSG-SRT.A.2**

SET 22 • HSG-CO.C.10

MAIN

Geometry main question • High School — Geometry • Geometry: Congruence • HSG-CO.C.10

Triangle ABC has A(0, 0), B(2a, 0), and C(2b, 2c), with a > 0 and c ≠ 0. D is the midpoint of AC and E is the midpoint of BC. Prove that DE ∥ AB and DE = ½AB.

Work space

Conclusion: _____

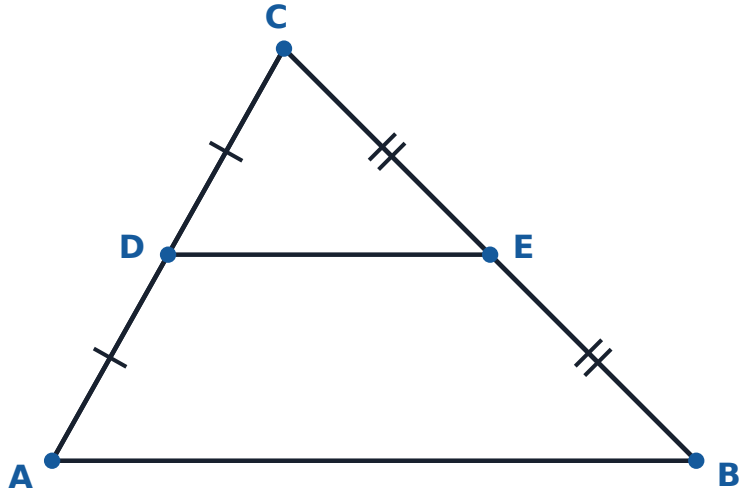
REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q1/5

In triangle ABC, D is the midpoint of AC and E is the midpoint of BC. Extend DE past E to F so that EF = DE. Prove that DE ∥ AB and DE = ½AB.

Work space



Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q2/5

In the coordinate proof of the midsegment theorem, the vertices are A(0, 0), B(2a, 0), C(2b, 2c). Why are the coordinates written with 2a, 2b, and 2c?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q3/5

A proof of the midsegment theorem with A(0, 0), B(2a, 0), C(2b, 2c):

1. D = (b, c), E = (a + b, c) (?)

2. slope DE = 0 = slope AB (?)

3. DE ∥ AB (?)

4. DE = a and AB = 2a, so DE = ½AB (?)

Give the four reasons.

Reasons: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q4/5

A student checks the triangle A(0, 0), B(6, 0), C(2, 4), finds the midsegment is parallel to AB and 3 units long, and concludes the theorem is proved. What is the flaw?

A. One example cannot prove a statement about every triangle

B. The midsegment should be 6 units long

C. Midpoints cannot be found with coordinates

D. The midsegment is not parallel to AB

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-M-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.10 • Set 22 • MAIN • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 1

Find midpoints of segments in the coordinate plane • Grade 6 • Geometry • 6.G.A.3

Find the midpoint of the segment from (2, 6) to (8, 2).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q1

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q1/5

Find the midpoint of the segment from $(-3, 1)$ to $(5, -7)$.

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q2

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q2/5

Find the midpoint of the segment from (0, 0) to (2a, 2c).

Midpoint: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q3

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q3/5

True or false?

The midpoint of the segment from (1, 1) to (7, 1) is (4, 1).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q4

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q4/5

True or false?

The midpoint of the segment from (0, 4) to (6, 0) is (6, 4).

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B1-Q5

Grade 6 • Geometry • 6.G.A.3 • Set 22 • BACKWARD 1 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 2

Find and compare slopes to decide whether lines are parallel • Grade 8 • Expressions & Equations • 8.EE.B.6

Find the slope of the line through (1, 2) and (4, 8).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q1

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q1/5

Find the slope of the line through (b, c) and (a + b, c).

Slope: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q2

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q2/5

True or false?

Two different nonvertical lines with the same slope are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q3

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q3/5

Which segment is parallel to the segment from (0, 0) to (4, 2)?

A. From (1, 3) to (5, 5)

B. From (2, 2) to (2, 6)

C. From (1, 3) to (3, 7)

D. From (0, 0) to (2, −1)

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q4

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q4/5

True or false?

A horizontal segment and a vertical segment have the same slope.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B2-Q5

Grade 8 • Expressions & Equations • 8.EE.B.6 • Set 22 • BACKWARD 2 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 3

Find lengths with the distance formula • Grade 8 • Geometry • 8.G.B.8

Find the distance between (b, c) and (a + b, c), with a > 0.

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q1

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q1/5

Find the distance between (0, 0) and (6, 8).

Distance: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q2

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q2/5

Find the length of the segment from $(-2, 5)$ to $(4, 5)$.

A. 3

B. 10

C. 6

D. 2

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q3

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q3/5

True or false?

The distance from (1, 2) to (4, 6) is 5.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q4

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q4/5

True or false?

The distance from (0, 0) to (2a, 0) is a, when $a > 0$.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B3-Q5

Grade 8 • Geometry • 8.G.B.8 • Set 22 • BACKWARD 3 • Q5/5

SET 22 • HSG-CO.C.10

BACKWARD 4

Prove lines parallel from congruent alternate interior angles • High School — Geometry • Geometry: Congruence • HSG-CO.C.9

Lines ℓ and m are cut by a transversal, and alternate interior angles are congruent. What can you conclude?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q1/5

True or false?

If two lines are cut by a transversal and alternate interior angles are congruent, the lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q2/5

In $\triangle ABC$, D is the midpoint of AC and E is the midpoint of BC. DE is extended to F so that $EF = DE$, and $\triangle DEC \cong \triangle FEB$. Which fact shows $BF \parallel AC$?

A. $DE = EF$

B. $\angle DCE \cong \angle FBE$ are alternate interior angles

C. $BF = DC$

D. E is the midpoint of BC

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q3/5

Two lines are cut by a transversal and corresponding angles measure 70° and 70°. Are the lines parallel?

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q4/5

True or false?

Congruent vertical angles prove that two lines are parallel.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-B4-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.9 • Set 22 • BACKWARD 4 • Q5/5

SET 22 • HSG-CO.C.10

FORWARD 1

Prove that the midpoints of a quadrilateral's sides form a parallelogram • High School — Geometry • Geometry: Congruence • HSG-CO.C.11

ABCD is any quadrilateral, and P, Q, R, S are the midpoints of AB, BC, CD, DA. Prove that PQRS is a parallelogram.

Work space

Conclusion: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q1

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q1/5

Quadrilateral ABCD has diagonals AC = 10 and BD = 14. Find the perimeter of the quadrilateral formed by the midpoints of its sides.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q2

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q2/5

True or false?

The midpoint quadrilateral of any quadrilateral is a parallelogram.

A. True

B. False

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q3

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q3/5

If the diagonals of ABCD are perpendicular, the midpoint quadrilateral is a

A. rectangle

B. trapezoid that is not a parallelogram

C. rhombus

D. kite that is not a parallelogram

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q4

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q4/5

In triangle ABC, D, E, F are the midpoints of AB, BC, CA. AB = 12, BC = 16, CA = 10. Find the perimeter of triangle DEF.

Perimeter: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F1-Q5

High School — Geometry • Geometry: Congruence • HSG-CO.C.11 • Set 22 • FORWARD 1 • Q5/5

SET 22 • HSG-CO.C.10

FORWARD 2

Midsegment triangles: similarity, scale factor, and area • High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2

D, E, F are the midpoints of the sides of triangle ABC. Explain why triangle DEF is similar to triangle ABC, and give the scale factor.

Answer: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q1

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q1/5

Triangle ABC has area 48. Find the area of the triangle formed by the midpoints of its sides.

Area: _____

REFERENCE LABEL — NOT PART OF THE STUDENT QUESTION

ID S22-F2-Q2

High School — Geometry • Geometry: Similarity, Right Triangles & Trigonometry • HSG-SRT.A.2 • Set 22 • FORWARD 2 • Q2/5